

CHAPTER 6

TIME-INDEPENDENT PERTURBATION THEORY

6.1 NONDEGENERATE PERTURBATION THEORY

6.1.1 General Formulation

Suppose we have solved the (time-independent) Schrödinger equation for some potential (say, the one-dimensional infinite square well):

$$H^0 \psi_n^0 = E_n^0 \psi_n^0, \quad [6.1]$$

obtaining a complete set of orthonormal eigenfunctions, ψ_n^0 ,

$$\langle \psi_n^0 | \psi_m^0 \rangle = \delta_{nm}, \quad [6.2]$$

and the corresponding eigenvalues E_n^0 . Now we perturb the potential slightly (say, by putting a little bump in the bottom of the well—Figure 6.1). We'd *like* to solve for the new eigenfunctions and eigenvalues:

$$H \psi_n = E_n \psi_n, \quad [6.3]$$

but unless we are very lucky, we're unlikely to be able to solve the Schrödinger equation exactly, for this more complicated potential. **Perturbation theory** is a systematic procedure for obtaining *approximate* solutions to the perturbed problem by building on the known exact solutions to the *unperturbed* case.

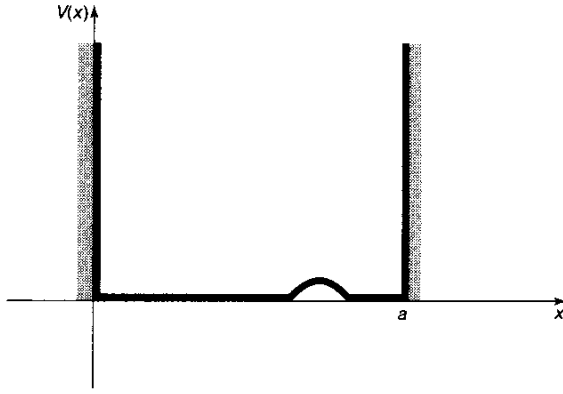


Figure 6.1: Infinite square well with small perturbation.

To begin with, we write the new Hamiltonian as the sum of two terms:

$$H = H^0 + \lambda H', \quad [6.4]$$

where H' is the perturbation. For the moment we'll take λ to be a small number; later we'll crank it up to 1, and H will be the true, exact Hamiltonian. Writing ψ_n and E_n as power series in λ , we have

$$\psi_n = \psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots; \quad [6.5]$$

$$E_n = E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots. \quad [6.6]$$

Here E_n^1 is the **first-order correction** to the n^{th} eigenvalue, and ψ_n^1 is the first-order correction to the n^{th} eigenfunction; E_n^2 and ψ_n^2 are the **second-order corrections**, and so on. Plugging Equations 6.4, 6.5, and 6.6 into Equation 6.3, we have

$$\begin{aligned} & (H^0 + \lambda H')[\psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots] \\ &= (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots)[\psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots], \end{aligned}$$

or (collecting like powers of λ):

$$\begin{aligned} & H^0 \psi_n^0 + \lambda(H^0 \psi_n^1 + H' \psi_n^0) + \lambda^2(H^0 \psi_n^2 + H' \psi_n^1) + \dots \\ &= E_n^0 \psi_n^0 + \lambda(E_n^0 \psi_n^1 + E_n^1 \psi_n^0) + \lambda^2(E_n^0 \psi_n^2 + E_n^1 \psi_n^1 + E_n^2 \psi_n^0) + \dots. \end{aligned}$$

To lowest order (λ^0) this yields $H^0 \psi_n^0 = E_n^0 \psi_n^0$, which is nothing new (just Equation 6.1). To first order (λ^1),

$$H^0 \psi_n^1 + H' \psi_n^0 = E_n^0 \psi_n^1 + E_n^1 \psi_n^0. \quad [6.7]$$

To second order (λ^2),

$$H^0 \psi_n^2 + H' \psi_n^1 = E_n^0 \psi_n^2 + E_n^1 \psi_n^1 + E_n^2 \psi_n^0, \quad [6.8]$$

and so on. (I'm done with λ , now—it was just a device to keep track of the different orders—so crank it up to 1.)

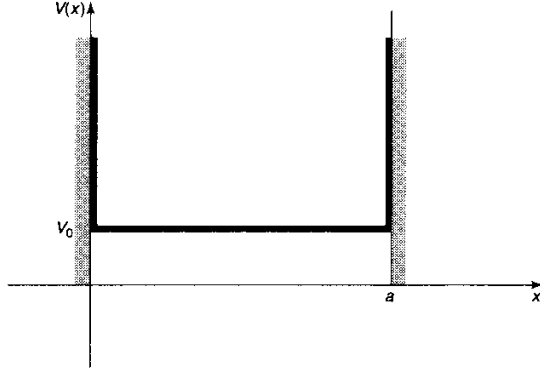


Figure 6.2: Constant perturbation over the whole well.

6.1.2 First-Order Theory

Taking the inner product of Equation 6.7 with ψ_n^0 [that is, multiplying by $(\psi_n^0)^*$ and integrating],

$$\langle \psi_n^0 | H^0 \psi_n^1 \rangle + \langle \psi_n^0 | H' \psi_n^0 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^1 \rangle + E_n^1 \langle \psi_n^0 | \psi_n^0 \rangle.$$

But H^0 is Hermitian, so

$$\langle \psi_n^0 | H^0 \psi_n^1 \rangle = \langle H^0 \psi_n^0 | \psi_n^1 \rangle = \langle E_n^0 \psi_n^0 | \psi_n^1 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^1 \rangle,$$

and this cancels the first term on the right. Moreover, $\langle \psi_n^0 | \psi_n^0 \rangle = 1$, so¹

$$E_n^1 = \langle \psi_n^0 | H' | \psi_n^0 \rangle. \quad [6.9]$$

This is the fundamental result of first-order perturbation theory; as a *practical* matter, it may well be the most important equation in quantum mechanics. It says that the first-order correction to the energy is the *expectation value* of the perturbation in the *unperturbed* state.

Example. The unperturbed wave functions for the infinite square well are (Equation 2.24)

$$\psi_n^0(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right).$$

Suppose first that we perturb the system by simply raising the “floor” of the well by a constant amount V_0 (Figure 6.2). In that case $H' = V_0$, and the first-order correction to the energy of the n^{th} state is

$$E_n^1 = \langle \psi_n^0 | V_0 | \psi_n^0 \rangle = V_0 \langle \psi_n^0 | \psi_n^0 \rangle = V_0.$$

¹In this context it doesn’t matter whether we write $\langle \psi_n^0 | H' \psi_n^0 \rangle$ or $\langle \psi_n^0 | H' | \psi_n^0 \rangle$ (with the extra vertical bar) because we are using the wave function itself to “label” the state. But the latter notation is preferable because it frees us from this specific convention.

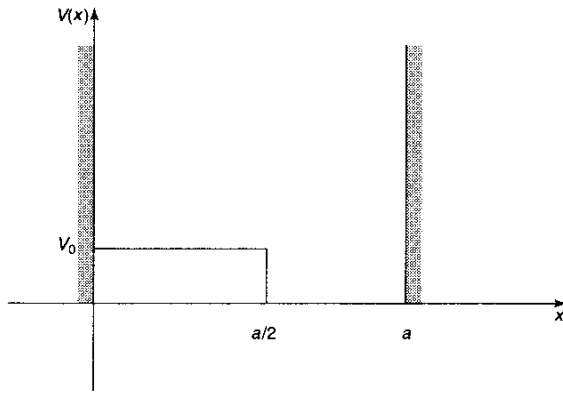


Figure 6.3: Constant perturbation over half the well.

The corrected energy levels, then, are $E_n \cong E_n^0 + V_0$; they are simply lifted by the amount V_0 . Of *course*! The only surprising thing is that in this case the first-order theory yields the *exact* answer. Evidently, for a *constant* perturbation all the higher corrections vanish.² If, on the other hand, the perturbation extends only halfway across the well (Figure 6.3), then

$$E_n^1 = \frac{2V_0}{a} \int_0^{a/2} \sin^2\left(\frac{n\pi}{a}x\right) dx = \frac{V_0}{2}.$$

In this case every energy level is lifted by $V_0/2$. That's not the *exact* result, presumably, but it does seem reasonable as a first-order approximation.

Equation 6.9 is the first-order correction to the *energy*; to find the first-order correction to the *wave function* we first rewrite Equation 6.7:

$$(H^0 - E_n^0)\psi_n^1 = -(H' - E_n^1)\psi_n^0. \quad [6.10]$$

The right side is a known function, so this amounts to an inhomogeneous differential equation for ψ_n^1 . Now, the unperturbed wave functions constitute a complete set, so ψ_n^1 (like any other function) can be expressed as a linear combination of them:

$$\psi_n^1 = \sum_{m \neq n} c_m^{(n)} \psi_m^0. \quad [6.11]$$

[There is no need to include $m = n$ in the sum, for if ψ_n^1 satisfies Equation 6.10, so too does $(\psi_n^1 + \alpha\psi_n^0)$, for any constant α , and we can use this freedom to subtract off the ψ_n^0 term.³] If we could determine the coefficients $c_m^{(n)}$, we'd be done. Well, putting Equation 6.11 into Equation 6.10, and using the fact that the ψ_m^0 satisfies the

²Incidentally, nothing here depends on the specific nature of the infinite square well—the same result applies for *any* potential, when the perturbation is constant.

³Alternatively, a glance at Equation 6.5 reveals that any ψ_n^0 component in ψ_n^1 might as well be pulled out and combined with the first term. We are *only* concerned, for the moment, with solving the Schrödinger equation (Equation 6.3), and the ψ_n we get will not, in general, be normalized.

unperturbed Schrödinger equation (Equation 6.1), we have

$$\sum_{m \neq n} (E_m^0 - E_n^0) c_m^{(n)} \psi_m^0 = -(H' - E_n^1) \psi_n^0.$$

Taking the inner product with ψ_l^0 ,

$$\sum_{m \neq n} (E_m^0 - E_n^0) c_m^{(n)} \langle \psi_l^0 | \psi_m^0 \rangle = -\langle \psi_l^0 | H' | \psi_n^0 \rangle + E_n^1 \langle \psi_l^0 | \psi_n^0 \rangle.$$

If $l = n$, the left side is zero, and we recover Equation 6.9; if $l \neq n$, we get

$$(E_l^0 - E_n^0) c_l^{(n)} = -\langle \psi_l^0 | H' | \psi_n^0 \rangle,$$

or

$$c_m^{(n)} = \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{E_n^0 - E_m^0}, \quad [6.12]$$

so

$$\psi_n^1 = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{(E_n^0 - E_m^0)} \psi_m^0. \quad [6.13]$$

Notice that the denominator is safe, since there is no coefficient with $m = n$, as long as the unperturbed energy spectrum is nondegenerate. But if two different unperturbed states share the same energy, we're in serious trouble (we divided by zero to get Equation 6.12); in that case we need **degenerate perturbation theory**, which I'll come to in Section 6.2.

That completes first-order perturbation theory: E_n^1 is given by Equation 6.9, and ψ_n^1 is given by Equation 6.13. I should warn you that whereas perturbation theory often yields surprisingly accurate energies (that is, $E_n^0 + E_n^1$ is quite close to the exact value E_n), the wave functions are notoriously poor.

***Problem 6.1** Suppose we put a delta-function bump in the center of the infinite square well:

$$H' = \alpha \delta(x - a/2),$$

where α is a constant. Find the first-order correction to the allowed energies. Explain why the energies are not perturbed for even n .

***Problem 6.2** For the harmonic oscillator [$V(x) = (1/2)kx^2$], the allowed energies are

$$E_n = (n + 1/2)\hbar\omega, \quad (n = 0, 1, 2, \dots),$$

where $\omega = \sqrt{k/m}$ is the classical frequency. Now suppose the spring constant increases slightly: $k \rightarrow (1 + \epsilon)k$. (Perhaps we cool the spring, so it becomes less flexible.)

- (a) Find the *exact* new energies (trivial, in this case). Expand your formula as a power series in ϵ , up to second order.
- (b) Now calculate the first-order perturbation in the energy, using Equation 6.6. What is H' here? Compare your result with part (a). *Hint*: It is not necessary—in fact, it is not *permitted*—to calculate a single integral in doing this problem.

Problem 6.3 Two identical bosons are placed in an infinite square well (Equation 2.15). They interact weakly with one another, via the potential

$$V(x_1, x_2) = -aV_0\delta(x_1 - x_2)$$

(where V_0 is a constant with the dimensions of energy and a is the width of the well

- (a) First, ignoring the interaction between the particles, find the ground state and first excited state—both the wave functions and the associated energies.
- (b) Use first-order perturbation theory to calculate the effect of the particle-particle interaction on the ground and first excited state energies.

6.1.3 Second-Order Energies

Proceeding as before, we take the inner product of the *second-order* equation (Equation 6.8) with ψ_n^0 :

$$\langle \psi_n^0 | H^0 \psi_n^2 \rangle + \langle \psi_n^0 | H' \psi_n^1 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^2 \rangle + E_n^1 \langle \psi_n^0 | \psi_n^1 \rangle + E_n^2 \langle \psi_n^0 | \psi_n^0 \rangle.$$

Again, we exploit the Hermiticity of H^0 :

$$\langle \psi_n^0 | H^0 \psi_n^2 \rangle = \langle H^0 \psi_n^0 | \psi_n^2 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^2 \rangle,$$

so the first term on the left cancels the first term on the right. Meanwhile, $\langle \psi_n^0 | \psi_n^0 \rangle = 1$, and we are left with a formula for E_n^2 :

$$E_n^2 = \langle \psi_n^0 | H' | \psi_n^1 \rangle - E_n^1 \langle \psi_n^0 | \psi_n^1 \rangle.$$

But

$$\langle \psi_n^0 | \psi_n^1 \rangle = \sum_{m \neq n} c_m^{(n)} \langle \psi_n^0 | \psi_m^0 \rangle = 0,$$

so

$$E_n^2 = \langle \psi_n^0 | H' | \psi_n^1 \rangle = \sum_{m \neq n} c_m^{(n)} \langle \psi_n^0 | H' | \psi_m^0 \rangle = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle \langle \psi_n^0 | H' | \psi_m^0 \rangle}{E_n^0 - E_m^0},$$

or, finally,

$$E_n^2 = \sum_{m \neq n} \frac{|\langle \psi_m^0 | H' | \psi_n^0 \rangle|^2}{E_n^0 - E_m^0}. \quad [6.14]$$

This is the fundamental result of second-order perturbation theory. We could proceed to calculate the second-order correction to the wave function (ψ_n^2), the third-order correction to the energy, and so on, but in practice Equation 6.14 is ordinarily as high as it is useful to pursue this method.

****Problem 6.4**

- (a) Find the second-order correction to the energies (E_n^2) for the potential in Problem 6.1. *Note:* You can sum the series explicitly to obtain the result $-2m(\alpha/\pi\hbar n)^2$, for odd n .
- (b) Calculate the second-order correction to the ground-state energy (E_0^2) for the potential in Problem 6.2. Check that your result is consistent with the exact solution.

****Problem 6.5** Consider a charged particle in the one-dimensional harmonic oscillator potential. Suppose we turn on a weak electric field (E) so that the potential energy is shifted by an amount $H' = -qEx$.

- (a) Show that there is no first-order change in the energy levels, and calculate the second-order correction. *Hint:* See Problem 3.50.
 - (b) The Schrödinger equation can be solved exactly in this case by a change of variables: $x' \equiv x - (qE/m\omega^2)$. Find the exact energies, and show that they are consistent with the perturbation theory approximation.
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6.2 DEGENERATE PERTURBATION THEORY

If the unperturbed states are degenerate—that is, if two (or more) distinct states (ψ_a^0 and ψ_b^0) share the same energy—then ordinary perturbation theory fails: $c_a^{(b)}$ (Equation 6.12) and E_a^2 (Equation 6.14) blow up (unless, possibly, the numerator vanishes, $\langle\psi_a^0|H'|\psi_b^0\rangle = 0$ —a loophole that will be important to us later on). In the degenerate case, therefore, there is no reason to trust even the *first*-order correction to the energy (Equation 6.9), and we must look for some other way to handle the problem.

6.2.1 Twofold Degeneracy

Suppose that

$$H^0\psi_a^0 = E^0\psi_a^0, \quad H^0\psi_b^0 = E^0\psi_b^0, \quad \text{and} \quad \langle\psi_a^0|\psi_b^0\rangle = 0. \quad [6.15]$$

Note that any linear combination of these states,

$$\psi^0 = \alpha\psi_a^0 + \beta\psi_b^0, \quad [6.16]$$

is still an eigenstate of H^0 , with the same eigenvalue E^0 :

$$H^0\psi^0 = E^0\psi^0. \quad [6.17]$$

Typically, the perturbation (H') will “break” the degeneracy: As we increase λ (from 0 to 1), the common unperturbed energy E^0 splits into two (Figure 6.4).

The essential problem is this: When we turn *off* the perturbation, the “upper” state reduces down to *one* linear combination of ψ_a^0 and ψ_b^0 , and the “lower” state reduces to some *other* linear combination, but we don’t know a priori *what* these “good” linear combinations will be. For this reason we can’t even calculate the *first-order* energy (Equation 6.9) because we don’t know what unperturbed states to use.

For the moment, therefore, let’s just write the “good” unperturbed states in the general form (Equation 6.16), keeping α and β adjustable. We want to solve the Schrödinger equation,

$$H\psi = E\psi, \quad [6.18]$$

with $H = H^0 + \lambda H'$ and

$$E = E^0 + \lambda E^1 + \lambda^2 E^2 + \dots, \quad \psi = \psi^0 + \lambda \psi^1 + \lambda^2 \psi^2 + \dots. \quad [6.19]$$

Plugging these into Equation 6.18, and collecting like powers of λ , as before, we find

$$H^0\psi^0 + \lambda(H'\psi^0 + H^0\psi^1) + \dots = E^0\psi^0 + \lambda(E^1\psi^0 + E^0\psi^1) + \dots.$$

But $H^0\psi^0 = E^0\psi^0$ (Equation 6.17), so the first terms cancel; at order λ^1 we have

$$H^0\psi^1 + H'\psi^0 = E^0\psi^1 + E^1\psi^0. \quad [6.20]$$

Taking the inner product with ψ_a^0 :

$$\langle \psi_a^0 | H^0 \psi^1 \rangle + \langle \psi_a^0 | H' \psi^0 \rangle = E^0 \langle \psi_a^0 | \psi^1 \rangle + E^1 \langle \psi_a^0 | \psi^0 \rangle.$$

Because H^0 is Hermitian, the first term on the left cancels the first term on the right. Putting in Equation 6.16 and exploiting the orthonormality condition Equation 6.15, we obtain

$$\alpha \langle \psi_a^0 | H' | \psi_a^0 \rangle + \beta \langle \psi_a^0 | H' | \psi_b^0 \rangle = \alpha E^1,$$

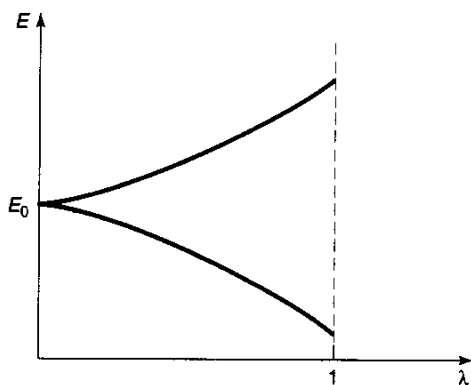


Figure 6.4: “Lifting” of a degeneracy by a perturbation.

or, more compactly,

$$\alpha W_{aa} + \beta W_{ab} = \alpha E^1, \quad [6.21]$$

where

$$W_{ij} \equiv \langle \psi_i^0 | H' | \psi_j^0 \rangle, \quad (i, j = a, b). \quad [6.22]$$

Similarly, the inner product with ψ_b^0 yields

$$\alpha W_{ba} + \beta W_{bb} = \beta E^1. \quad [6.23]$$

Notice that the W 's are (in principle) *known*—they are just the “matrix elements” of H' , with respect to the unperturbed wave functions ψ_a^0 and ψ_b^0 . Multiplying Equation 6.23 by W_{ab} , and using Equation 6.21 to eliminate βW_{ab} , we find

$$\alpha[W_{ab}W_{ba} - (E^1 - W_{aa})(E^1 - W_{bb})] = 0. \quad [6.24]$$

If α is *not* zero, Equation 6.24 yields an equation for E^1 :

$$(E^1)^2 - E^1(W_{aa} + W_{bb}) + (W_{aa}W_{bb} - W_{ab}W_{ba}) = 0. \quad [6.25]$$

Invoking the quadratic formula, and noting (from Equation 6.22) that $W_{ba} = W_{ab}^*$, we conclude that

$$E_{\pm}^1 = \frac{1}{2} \left[W_{aa} + W_{bb} \pm \sqrt{(W_{aa} - W_{bb})^2 + 4|W_{ab}|^2} \right]. \quad [6.26]$$

This is the fundamental result of degenerate perturbation theory; the two roots correspond to the two perturbed energies.

But what if α is zero? In that case $\beta = 1$, Equation 6.21 says $W_{ab} = 0$, and Equation 6.23 gives $E^1 = W_{bb}$. This is actually included in the general result (Equation 6.26), with the plus sign (the minus sign corresponds to $\alpha = 1$, $\beta = 0$). What's more, the *answers*,

$$E_+^1 = W_{bb} = \langle \psi_b^0 | H' | \psi_b^0 \rangle, \quad E_-^1 = W_{aa} = \langle \psi_a^0 | H' | \psi_a^0 \rangle,$$

are precisely what we would have obtained using *nondegenerate* perturbation theory (Equation 6.9)—we have simply been *lucky*: The states ψ_a^0 and ψ_b^0 were *already* the “correct” linear combinations. Obviously, it would be greatly to our advantage if we could somehow *guess* the “good” states right from the start. In practice, we can often do so by exploiting the following theorem:

Theorem: Let A be a Hermitian operator that commutes with H' . If ψ_a^0 and ψ_b^0 are eigenfunctions of A with distinct eigenvalues,

$$A\psi_a^0 = \mu\psi_a^0, \quad A\psi_b^0 = \nu\psi_b^0, \quad \text{and } \mu \neq \nu,$$

then $W_{ab} = 0$ (and hence ψ_a^0 and ψ_b^0 are the “good” states to use in perturbation theory).

Proof: By assumption, $[A, H'] = 0$, so

$$\begin{aligned}\langle \psi_a^0 | [A, H'] \psi_b^0 \rangle &= 0 \\ &= \langle \psi_a^0 | A H' \psi_b^0 \rangle - \langle \psi_a^0 | H' A \psi_b^0 \rangle \\ &= \langle A \psi_a^0 | H' \psi_b^0 \rangle - \langle \psi_a^0 | H' \nu \psi_b^0 \rangle \\ &= (\mu - \nu) \langle \psi_a^0 | H' \psi_b^0 \rangle = (\mu - \nu) W_{ab}.\end{aligned}$$

But $\mu \neq \nu$, so $W_{ab} = 0$. QED

Moral: If you're faced with degenerate states, look around for some Hermitian operator A that commutes with H' ; pick as your unperturbed states ones that are simultaneously eigenfunctions of H^0 and A . Then use *ordinary* first-order perturbation theory. If you can't find such an operator, you'll have to resort to Equation 6.26, but in practice this is seldom necessary.

Problem 6.6 Let the two "good" unperturbed states be

$$\psi_{\pm}^0 = \alpha_{\pm} \psi_a^0 + \beta_{\pm} \psi_b^0,$$

where $\alpha_{\pm}$ and $\beta_{\pm}$ are determined (up to normalization) by Equation 6.21 (or Equation 6.23), with Equation 6.26 for $E_{\pm}$. Show explicitly that

- (a) $\psi_{\pm}^0$ are orthogonal ($\langle \psi_+^0 | \psi_-^0 \rangle = 0$);
- (b) $\langle \psi_+^0 | H' | \psi_-^0 \rangle = 0$;
- (c) $\langle \psi_{\pm}^0 | H' | \psi_{\pm}^0 \rangle = E_{\pm}^1$.

Problem 6.7 Consider a particle of mass m that is free to move in a one-dimensional region of length L that closes on itself (for instance, a bead which slides frictionlessly on a circular wire of circumference L ; Problem 2.43).

- (a) Show that the stationary states can be written in the form

$$\psi_n(x) = \frac{1}{\sqrt{L}} e^{2\pi i n x / L}, \quad (-L/2 < x < L/2),$$

where $n = 0, \pm 1, \pm 2, \dots$, and the allowed energies are

$$E_n = \frac{2}{m} \left(\frac{n\pi\hbar}{L} \right)^2.$$

Notice that—with the exception of the ground state ($n = 0$)—these are all doubly degenerate.

- (b) Now suppose we introduce the perturbation

$$H' = -V_0 e^{-x^2/a^2},$$

where $a \ll L$. (This puts a little “dimple” in the potential at $x = 0$, as though we bent the wire slightly to make a “trap.”) Find the first-order correction to E_n , using Equation 6.26. *Hint:* To evaluate the integrals, exploit the fact that $a \ll L$ to extend the limits from $\pm L/2$ to $\pm\infty$; after all, H' is essentially zero outside $-a < x < a$.

- (c) What are the “good” linear combinations of ψ_n and ψ_{-n} for this problem? Show that with these states you get the first-order correction using Equation 6.9.
- (d) Find a Hermitian operator A that fits the requirements of the theorem, and show that the simultaneous eigenstates of H^0 and A are precisely the ones you found in (c).

6.2.2 Higher-Order Degeneracy

In the previous section I assumed the degeneracy was twofold, but it is easy to see how the method generalizes. Rewrite Equations 6.21 and 6.23 in matrix form:

$$\begin{pmatrix} W_{aa} & W_{ab} \\ W_{ba} & W_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E^1 \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \quad [6.27]$$

Evidently the E^1 's are nothing but the *eigenvalues* of the W -matrix; Equation 6.25 is the characteristic equation (Equation 3.70) for this matrix, and the “good” linear combinations of the unperturbed states are the eigenvectors of W . In the case of n -fold degeneracy, we look for the eigenvalues of the $n \times n$ matrix

$$W_{ij} = \langle \psi_i^0 | H' | \psi_j^0 \rangle. \quad [6.28]$$

In the language of linear algebra, finding the “good” unperturbed wave functions amounts to constructing a basis in the degenerate subspace that *diagonalizes* the perturbation H' . Once again, if you can think of an operator A that *commutes* with H' , and use the simultaneous eigenfunctions of A and H^0 , then the W matrix will *automatically* be diagonal, and you won't have to fuss with solving the characteristic equation.

Example. Consider the three-dimensional infinite cubical well (Problem 4.2):

$$V(x, y, z) = \begin{cases} 0, & \text{if } 0 < x < a, 0 < y < a, \text{ and } 0 < z < a; \\ \infty & \text{otherwise.} \end{cases} \quad [6.29]$$

The stationary states are

$$\psi_{n_x n_y n_z}^0(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{a} y\right) \sin\left(\frac{n_z \pi}{a} z\right), \quad [6.30]$$

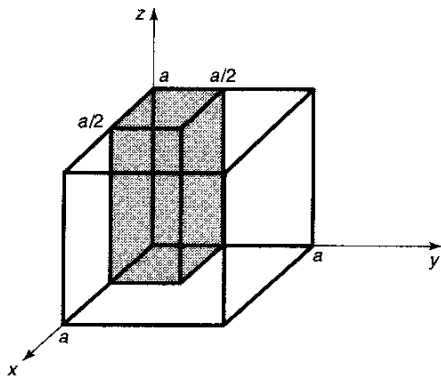


Figure 6.5: The perturbation increases the potential by an amount V_0 in the shaded sector.

where n_x , n_y , and n_z are positive integers. The corresponding allowed energies are

$$E_{n_x n_y n_z}^0 = \frac{\pi^2 \hbar^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2). \quad [6.31]$$

Notice that the ground state (ψ_{111}) is nondegenerate; its energy is

$$E_0^0 \equiv 3 \frac{\pi^2 \hbar^2}{2ma^2}. \quad [6.32]$$

But the first excited state is (triply) degenerate:

$$\psi_a \equiv \psi_{112}, \quad \psi_b \equiv \psi_{121}, \quad \text{and} \quad \psi_c \equiv \psi_{211}, \quad [6.33]$$

all share the energy

$$E_1^0 \equiv 3 \frac{\pi^2 \hbar^2}{ma^2}. \quad [6.34]$$

Now let's introduce the perturbation

$$H' = \begin{cases} V_0, & \text{if } 0 < x < a/2 \text{ and } 0 < y < a/2; \\ 0, & \text{otherwise.} \end{cases} \quad [6.35]$$

This raises the potential by an amount V_0 in one quarter of the box (see Figure 6.5). The first-order correction to the ground state energy is given by Equation 6.9:

$$\begin{aligned} E_0^1 &= \langle \psi_{111} | H' | \psi_{111} \rangle = \left(\frac{2}{a} \right)^3 V_0 \int_0^{a/2} \sin^2 \left(\frac{\pi}{a} x \right) dx \\ &\quad \int_0^{a/2} \sin^2 \left(\frac{\pi}{a} y \right) dy \int_0^a \sin^2 \left(\frac{\pi}{a} z \right) dz = \frac{1}{4} V_0, \end{aligned} \quad [6.36]$$

which is just what we would expect.

For the first excited state we need the full machinery of degenerate perturbation theory. The first step is to construct the matrix W . The diagonal elements are the

same as for the ground state (except that the argument of one of the sines is doubled); you can check for yourself that

$$W_{aa} = W_{bb} = W_{cc} = \frac{1}{4}V_0.$$

The off-diagonal elements are more interesting:

$$\begin{aligned} W_{ab} = & \left(\frac{2}{a}\right)^3 V_0 \int_0^{a/2} \sin^2\left(\frac{\pi}{a}x\right) dx \\ & \int_0^{a/2} \sin\left(\frac{\pi}{a}y\right) \sin\left(\frac{2\pi}{a}y\right) dy \int_0^a \sin\left(\frac{2\pi}{a}z\right) \sin\left(\frac{\pi}{a}z\right) dz. \end{aligned}$$

But the z integral is zero (as it will be also for W_{ac}), so

$$W_{ab} = W_{ac} = 0.$$

Finally,

$$\begin{aligned} W_{bc} = & \left(\frac{2}{a}\right)^3 V_0 \int_0^{a/2} \sin\left(\frac{\pi}{a}x\right) \sin\left(\frac{2\pi}{a}x\right) dx \\ & \int_0^{a/2} \sin\left(\frac{2\pi}{a}y\right) \sin\left(\frac{\pi}{a}y\right) dy \int_0^a \sin^2\left(\frac{\pi}{a}z\right) dz = \frac{16}{9\pi^2}V_0. \end{aligned}$$

Thus

$$W = \frac{V_0}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \kappa \\ 0 & \kappa & 1 \end{pmatrix}, \quad [6.37]$$

where $\kappa \equiv (8/3\pi)^2 \approx 0.7205$.

The characteristic equation for W (or rather, for $4W/V_0$, which is easier to work with) is

$$(1 - w)^3 - \kappa^2(1 - w) = 0,$$

and the eigenvalues are

$$w_1 = 1; \quad w_2 = 1 + \kappa \approx 1.7205; \quad w_3 = 1 - \kappa \approx 0.2795.$$

To first order in λ , then,

$$E_1(\lambda) = \begin{cases} E_1^0 + \lambda V_0/4, \\ E_1^0 + \lambda(1 + \kappa)V_0/4, \\ E_1^0 + \lambda(1 - \kappa)V_0/4, \end{cases} \quad [6.38]$$

where E_1^0 is the (common) unperturbed energy (Equation 6.34). The perturbation lifts the degeneracy, splitting E_1^0 into three distinct energy levels (see Figure 6.6). Notice that if we had naively applied *nondegenerate* perturbation theory to this problem, we would have concluded that the first-order correction (Equation 6.9) is the

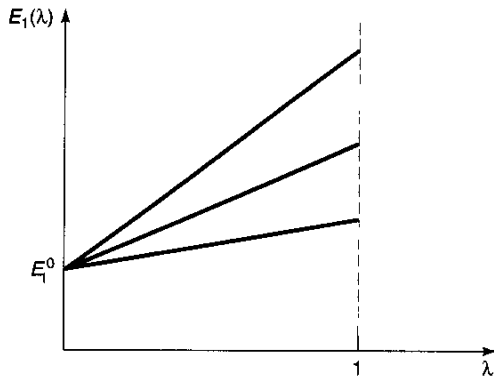


Figure 6.6: Lifting of the degeneracy in the example (Equation 6.38).

same for all three states, and equal to $V_0/4$ —which is actually correct only for the middle state.

Meanwhile, the “good” unperturbed states are linear combinations of the form

$$\psi^0 = \alpha\psi_a + \beta\psi_b + \gamma\psi_c, \quad [6.39]$$

where the coefficients (α , β , and γ) form the eigenvectors of the matrix W :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \kappa \\ 0 & \kappa & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = w \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}.$$

For $w = 1$ we get $\alpha = 1$, $\beta = \gamma = 0$; for $w = 1 \pm \kappa$ we get $\alpha = 0$, $\beta = \pm\gamma = 1/\sqrt{2}$. (I normalized them as I went along.) Thus the “good” states are⁴

$$\psi^0 = \begin{cases} \psi_a, \\ (\psi_b + \psi_c)/\sqrt{2}, \\ (\psi_b - \psi_c)/\sqrt{2}. \end{cases} \quad [6.40]$$

Problem 6.8 Suppose we perturb the infinite cubical well (Equation 6.29) by putting a delta-function “bump” at the point $(a/4, a/2, 3a/4)$:

$$H' = a^3 V_0 \delta(x - a/4) \delta(y - a/2) \delta(z - 3a/4).$$

Find the first-order corrections to the energy of the ground state and the (triply degenerate) first excited states.

⁴We might have guessed this result right from the start by noting that the operator P_{xy} , which interchanges x and y , commutes with H' . Its eigenvalues are $+1$ (for functions that are *even* under the interchange) and -1 (for functions that are *odd*). In this case ψ_a is *already* even, $(\psi_b + \psi_c)$ is even, and $(\psi_b - \psi_c)$ is odd. However, this is not quite conclusive, since any linear combination of the even states would still be even. What we’d *really* like is an operator with *three* distinct eigenvalues in the degenerate subspace.

***Problem 6.9** Consider a quantum system with just *three* linearly independent states. The Hamiltonian, in matrix form, is

$$H = V_0 \begin{pmatrix} (1 - \epsilon) & 0 & 0 \\ 0 & 1 & \epsilon \\ 0 & \epsilon & 2 \end{pmatrix},$$

where V_0 is a constant and ϵ is some small number ($\epsilon \ll 1$).

- (a) Write down the eigenvectors and eigenvalues of the *unperturbed* Hamiltonian ($\epsilon = 0$).
- (b) Solve for the *exact* eigenvalues of H . Expand each of them as a power series in ϵ , up to second order.
- (c) Use first- and second-order *nondegenerate* perturbation theory to find the approximate eigenvalue for the state that grows out of the nondegenerate eigenvector of H^0 . Compare the exact result from (b).
- (d) Use *degenerate* perturbation theory to find the first-order correction to the two initially degenerate eigenvalues. Compare the exact results.

6.3 THE FINE STRUCTURE OF HYDROGEN

In our study of the hydrogen atom (Section 4.2), we took the Hamiltonian to be

$$H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \quad [6.41]$$

(electron kinetic energy plus Coulombic potential energy). But this is not quite the whole story. We have already learned how to correct for the motion of the nucleus: Just replace m by the reduced mass (Problem 5.1). More significant is the so-called **fine structure**, which is actually due to two distinct mechanisms: a **relativistic correction**, and **spin-orbit coupling**. Compared to the Bohr energies (Equation 4.70), fine structure is a tiny perturbation—smaller by a factor of α^2 , where

$$\alpha \equiv \frac{e^2}{4\pi\epsilon_0\hbar c} \cong \frac{1}{137.036} \quad [6.42]$$

is the famous **fine structure constant**. Smaller still (by another factor of α) is the **Lamb shift**, associated with the quantization of the Coulomb field, and smaller by yet another order of magnitude is the **hyperfine structure**, which is due to the magnetic interaction between the dipole moments of the electron and the proton. This hierarchy is summarized in Table 6.1.

Table 6.1: Hierarchy of corrections to the Bohr energies of hydrogen.

Bohr energies:	of order	$\alpha^2 mc^2$
Fine structure:	of order	$\alpha^4 mc^2$
Lamb shift:	of order	$\alpha^5 mc^2$
Hyperfine splitting:	of order	$(m/m_p)\alpha^4 mc^2$

In the present section we will analyze the fine structure of hydrogen, as an application of time-independent perturbation theory.

Problem 6.10

- (a) Express the Bohr energies in terms of the fine structure constant and the rest energy (mc^2) of the electron.
- (b) Calculate the fine structure constant from first principles (i.e. without recourse to the empirical values of ϵ_0 , e , $\hbar$, and c). *Note:* The fine structure constant is undoubtedly the most fundamental pure (dimensionless) number in all of physics: It relates the basic constants of electromagnetism (the charge of the electron), relativity (the speed of light), and quantum mechanics (Planck's constant). If you can solve part (b), you have the most certain Nobel Prize in history waiting for you. But I wouldn't recommend spending a lot of time on it right now; many smart people have tried and given up.

6.3.1 The Relativistic Correction

The first term in the Hamiltonian is supposed to represent kinetic energy:

$$T = \frac{1}{2}mv^2 = \frac{p^2}{2m}, \quad [6.43]$$

and the canonical substitution $\mathbf{p} \rightarrow (\hbar/i)\nabla$ yields the operator

$$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2. \quad [6.44]$$

But Equation 6.43 is the *classical* equation for kinetic energy; the *relativistic* formula is

$$T = \frac{mc^2}{\sqrt{1 - (v/c)^2}} - mc^2. \quad [6.45]$$

The first term is the *total* relativistic energy (not counting *potential* energy, which we aren't concerned with at the moment), and the second term is the *rest* energy—the *difference* is the energy attributable to motion. We need to express T in terms of the

(relativistic) momentum,

$$p = \frac{mv}{\sqrt{1 - (v/c)^2}}, \quad [6.46]$$

instead of velocity. Notice that

$$p^2 c^2 + m^2 c^4 = \frac{m^2 v^2 c^2 + m^2 c^4 [1 - (v/c)^2]}{1 - (v/c)^2} = \frac{m^2 c^4}{1 - (v/c)^2} = (T + mc^2)^2,$$

so

$$T = \sqrt{p^2 c^2 + m^2 c^4} - mc^2. \quad [6.47]$$

This relativistic equation for kinetic energy reduces (of course) to the classical result (Equation 6.43), in the nonrelativistic limit $p \ll mc$; expanding in powers of the small number (p/mc) , we have

$$\begin{aligned} T &= mc^2 \left[\sqrt{1 + \left(\frac{p}{mc}\right)^2} - 1 \right] = mc^2 \left[1 + \frac{1}{2} \left(\frac{p}{mc}\right)^2 - \frac{1}{8} \left(\frac{p}{mc}\right)^4 \cdots - 1 \right] \\ &= \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \cdots. \end{aligned} \quad [6.48]$$

The lowest-order⁵ relativistic contribution to the Hamiltonian is evidently

$$H'_r = -\frac{\hat{p}^4}{8m^3 c^2}. \quad [6.49]$$

In first-order perturbation theory, the correction to E_n is given by the expectation value of H' in the unperturbed state (Equation 6.9):

$$E_r^1 = \langle H'_r \rangle = -\frac{1}{8m^3 c^2} \langle \psi | \hat{p}^4 | \psi \rangle = -\frac{1}{8m^3 c^2} \langle \hat{p}^2 \psi | \hat{p}^2 \psi \rangle. \quad [6.50]$$

Now the Schrödinger equation (for the unperturbed states) says

$$\hat{p}^2 \psi = 2m(E - V)\psi, \quad [6.51]$$

and hence⁶

$$E_r^1 = -\frac{1}{2mc^2} \langle (E - V)^2 \rangle = -\frac{1}{2mc^2} [E^2 - 2E\langle V \rangle + \langle V^2 \rangle]. \quad [6.52]$$

⁵The kinetic energy of the electron in hydrogen is on the order of 10 eV, which is miniscule compared to its rest energy (511,000 eV), so the hydrogen atom is basically nonrelativistic, and we can afford to keep only the lowest-order correction. In Equation 6.48, p is the *relativistic* momentum (Equation 6.46), *not* the classical momentum mv . It is the former that we now associate with the quantum operator $-i\hbar\nabla$, in Equation 6.49.

⁶There is some sleight-of-hand in this maneuver, which exploits the Hermiticity of $\hat{p}^2$ and of $(E - V)$. In truth, the operator $\hat{p}^4$ is not Hermitian, for states with $l = 0$, and the applicability of perturbation theory to Equation 6.49 is therefore called into question. Fortunately, the *exact* solution is available; it can be obtained by using the (relativistic) Dirac equation in place of the (nonrelativistic) Schrödinger equation, and it confirms the results we obtain here by less rigorous means. (See Problem 6.17.)

So far, this is entirely general; but we're interested in the case of hydrogen, for which $V(r) = -(1/4\pi\epsilon_0)e^2/r$:

$$E_r^1 = -\frac{1}{2mc^2} \left[E_n^2 + 2E_n \left(\frac{e^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle + \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left\langle \frac{1}{r^2} \right\rangle \right], \quad [6.53]$$

where E_n is the Bohr energy of the state in question.

To complete the job, we need the expectation values of $1/r$ and $1/r^2$ in the (unperturbed) state ψ_{nlm} (Equation 4.89). The first is easy (see Problem 6.11):

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a}, \quad [6.54]$$

where a is the Bohr radius (Equation 4.72). The second is not so simple to derive (see Problem 6.28), but the answer is⁷

$$\left\langle \frac{1}{r^2} \right\rangle = \frac{1}{(l + 1/2)n^3 a^2}. \quad [6.55]$$

It follows that

$$E_r^1 = -\frac{1}{2mc^2} \left[E_n^2 + 2E_n \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2 a} + \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{(l + 1/2)n^3 a^2} \right],$$

or, eliminating a (using Equation 4.72) and expressing everything in terms of E_n (using Equation 4.70),

$$E_r^1 = -\frac{E_n^2}{2mc^2} \left[\frac{4n}{l + 1/2} - 3 \right]. \quad [6.56]$$

Notice that the relativistic correction is smaller than E_n by a factor of $E_n/mc^2 \approx 2 \times 10^{-5}$.

You might have noticed that I used *nondegenerate* perturbation theory in this calculation even though the hydrogen atom is highly degenerate. But the perturbation is spherically symmetrical, so it commutes with L^2 and L_z . Moreover, the eigenfunctions of these operators (taken together) have distinct eigenvalues for the n^2 states with a given E_n . Luckily, then, the wave functions ψ_{nlm} are “good” states for this problem, so as it happens the use of nondegenerate perturbation theory was legitimate.

***Problem 6.11** Use the virial theorem (Problem 4.41) to prove Equation 6.54.

⁷The general formula for the expectation value of any power of r is given in Hans A. Bethe and Edwin E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*, (New York: Plenum, 1977), p. 17.

Problem 6.12 In Problem 4.43, you calculated the expectation value of r^s in the state ψ_{321} . Check your answer for the special cases $s = 0$ (trivial), $s = -1$ (Equation 6.54), $s = -2$ (Equation 6.55), and $s = -3$ (Equation 6.63). Comment on the case $s = -7$.

****Problem 6.13** Find the (lowest-order) relativistic correction to the energy levels of the one-dimensional harmonic oscillator. *Hint:* Use the technique of Problem 2.37.

6.3.2 Spin-Orbit Coupling

Imagine the electron in orbit around the nucleus; from the *electron's* point of view, the proton is circling around *it* (Figure 6.7). This orbiting positive charge sets up a magnetic field $\mathbf{B}$ in the electron frame, which exerts a torque on the spinning electron, tending to align its magnetic moment ($\boldsymbol{\mu}$) along the direction of the field. The Hamiltonian (Equation 4.157) is

$$H = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad [6.57]$$

The Magnetic Field of the Proton. If we picture the proton (from the electron's perspective) as a continuous current loop (Figure 6.7), its magnetic field can be calculated from the Biot-Savart law:

$$B = \frac{\mu_0 I}{2r},$$

with an effective current $I = e/T$, where e is the charge of the proton and T is the period of the orbit. On the other hand, the orbital angular momentum of the *electron* (in the rest frame of the *nucleus*) is $L = rmv = 2\pi mr^2/T$. Moreover, $\mathbf{B}$ and $\mathbf{L}$ point in the same direction (up, in Figure 6.7), so

$$\mathbf{B} = \frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} \mathbf{L}. \quad [6.58]$$

(I used $c = 1/\sqrt{\epsilon_0\mu_0}$ to eliminate μ_0 in favor of ϵ_0 .)

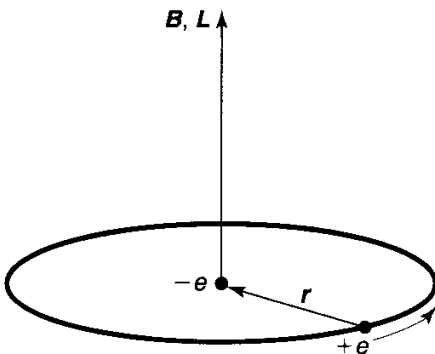


Figure 6.7: Hydrogen atom, from the electron's perspective.

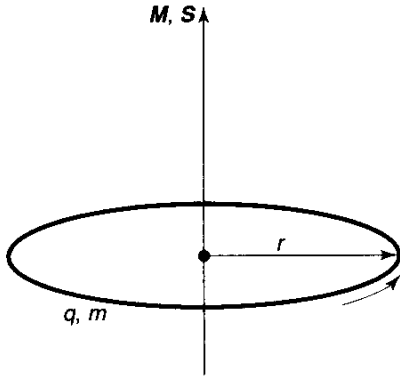


Figure 6.8: A ring of charge, rotating about its axis.

The Magnetic Dipole Moment of the Electron. The magnetic dipole moment of a spinning charge is related to its (spin) angular momentum; the proportionality factor is the gyromagnetic ratio (which we already encountered in Section 4.4.2). Let's derive it, using classical electrodynamics. Consider first a charge q smeared out around a ring of radius r , which rotates about the axis with period T (Figure 6.8). The magnetic dipole moment of the ring is defined as the current (q/T) times the area (πr^2):

$$\mu = \frac{q\pi r^2}{T}.$$

If the mass of the ring is m , its angular momentum is the moment of inertia (mr^2) times the angular velocity ($2\pi/T$):

$$S = \frac{2\pi mr^2}{T}.$$

The gyromagnetic ratio for this configuration is evidently $\mu/S = q/2m$. Notice that it is independent of r (and T). If I had some more complicated object, such as a sphere (all I require is that it be a figure of revolution, rotating about its axis), I could calculate μ and S by chopping it into little rings and adding their contributions. As long as the mass and the charge are distributed in the same manner (so that the charge-to-mass ratio is uniform), the gyromagnetic ratio will be the same for each ring, and hence also for the object as a whole. Moreover, the directions of μ and S are the same (or opposite, if the charge is negative), so

$$\mu = \left(\frac{q}{2m}\right) S.$$

That was a purely *classical* calculation, however; as it turns out, the electron's magnetic moment is *twice* the classical answer:

$$\mu_e = -\frac{e}{m} S. \quad [6.59]$$

The “extra” factor of 2 was explained by Dirac in his relativistic theory of the electron.⁸

The Spin-Orbit Interaction. Putting all this together, we have

$$H = \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{m^2 c^2 r^3} \mathbf{S} \cdot \mathbf{L}.$$

But there is a serious fraud in this calculation: I did the analysis in the rest frame of the electron, but that’s *not an inertial system*—it *accelerates* as the electron orbits around the nucleus. You can get away with this if you make an appropriate kinematic correction, known as the **Thomas precession**.⁹ In this context it throws in a factor of 1/2:

$$H'_{\text{so}} = \left(\frac{e^2}{8\pi\epsilon_0} \right) \frac{1}{m^2 c^2 r^3} \mathbf{S} \cdot \mathbf{L}. \quad [6.60]$$

This is the spin-orbit interaction; apart from two corrections (the modified gyromagnetic ratio for the electron and the Thomas precession factor—which, coincidentally, exactly cancel one another), it is just what you would expect on the basis of a naive classical model. Physically, it is attributable to the torque exerted on the magnetic dipole moment of the spinning electron, by the magnetic field of the proton, in the electron’s instantaneous rest frame.

Now the quantum mechanics. In the presence of spin-orbit coupling, the Hamiltonian no longer commutes with $\mathbf{L}$ and $\mathbf{S}$, so the spin and orbital angular momenta are not separately conserved (see Problem 6.14). However, H'_{so} *does* commute with L^2 , S^2 , and the *total* angular momentum

$$\mathbf{J} \equiv \mathbf{L} + \mathbf{S}, \quad [6.61]$$

and hence these quantities *are* conserved (Equation 3.148). To put it another way, the eigenstates of L_z and S_z are not “good” states to use in perturbation theory, but the eigenstates of L^2 , S^2 , J^2 , and J_z are. Now

$$J^2 = (\mathbf{L} + \mathbf{S}) \cdot (\mathbf{L} + \mathbf{S}) = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S},$$

⁸We have already noticed that it can be dangerous to picture the electron as a spinning sphere (see Problem 4.26), and it is not too surprising that this naive classical model gets the gyromagnetic ratio wrong. Incidentally, quantum electrodynamics reveals tiny corrections to Equation 6.59; the calculation of the so-called **anomalous magnetic moment** of the electron stands as one of the greatest achievements of theoretical physics.

⁹One way of thinking of it is that we are continually stepping from one inertial system to another; Thomas precession amounts to the cumulative effect of all these Lorentz transformations. We could avoid this problem, of course, by staying in the *lab* frame, in which the nucleus is at rest. In that case the field of the proton is purely *electric*, and you may well wonder why it exerts any torque on the electron. Well, the fact is that a moving *magnetic* dipole acquires an *electric* dipole moment, and in the lab frame the spin-orbit coupling is due to the interaction of the *electric* field of the nucleus with the *electric* dipole moment of the electron. Because this analysis requires more sophisticated electrodynamics, it seems best to adopt the electron’s perspective, where the physical mechanism is more transparent. For a related discussion, see V. Namias, *Am. J. Phys.*, **57**, 171 (1989).

so

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2), \quad [6.62]$$

and therefore the eigenvalues of $\mathbf{L} \cdot \mathbf{S}$ are

$$\frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)].$$

In this case, of course, $s = 1/2$. Meanwhile, the expectation value of $1/r^3$ (see Problem 6.30) is

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{l(l+1/2)(l+1)n^3a^3}, \quad [6.63]$$

and we conclude that

$$E_{\text{so}}^1 = \langle H'_{\text{so}} \rangle = \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2c^2} \frac{(\hbar^2/2)[j(j+1) - l(l+1) - 3/4]}{l(l+1/2)(l+1)n^3a^3},$$

or, expressing it all in terms of E_n :

$$E_{\text{so}}^1 = \frac{E_n^2}{mc^2} \left\{ \frac{n[j(j+1) - l(l+1) - 3/4]}{l(l+1/2)(l+1)} \right\}. \quad [6.64]$$

It is remarkable, considering the totally different physical mechanisms involved, that the relativistic correction and the spin-orbit coupling are of the same order (E_n^2/mc^2). Adding them together, we get the complete fine-structure formula (see Problem 6.15):

$$E_{\text{fs}}^1 = \frac{E_n^2}{2mc^2} \left(3 - \frac{4n}{j+1/2} \right). \quad [6.65]$$

Combining this with the Bohr formula, we obtain the grand result for the energy levels of hydrogen, including fine structure:

$$E_{nj} = -\frac{13.6 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right) \right]. \quad [6.66]$$

Fine structure breaks the degeneracy in l (that is, for a given n , the different allowed values of l do not all carry the same energy); the energies are determined by n and j (see Figure 6.9). The azimuthal eigenvalues for orbital and spin angular momentum (m_l and m_s) are no longer “good” quantum numbers—the stationary states are linear combinations of states with different values of these quantities; the “good” quantum numbers are n , l , s , j , and m_j .¹⁰

¹⁰To write $|jm_j\rangle$ (for given l and s) as a linear combination of $|lm_l\rangle|sm_s\rangle$ we would use the appropriate Clebsch-Gordan coefficients (Equation 4.185).

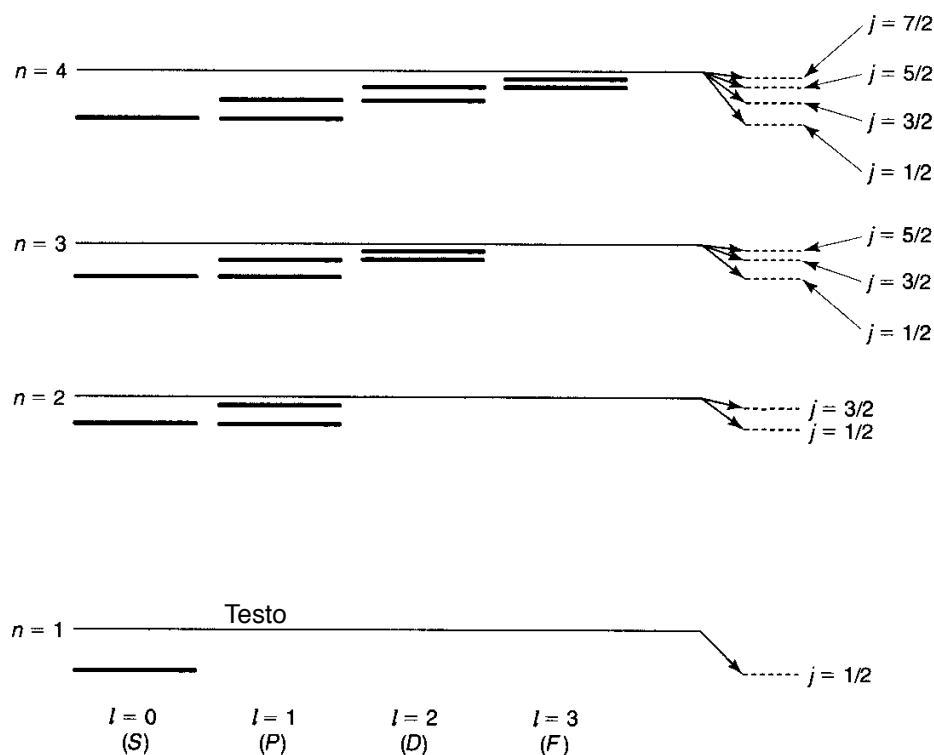


Figure 6.9: Energy levels of hydrogen, including fine structure (not to scale).

Problem 6.14 Evaluate the following commutators: (a) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{L}]$, (b) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{S}]$, (c) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{J}]$, (d) $[\mathbf{L} \cdot \mathbf{S}, L^2]$, (e) $[\mathbf{L} \cdot \mathbf{S}, S^2]$, (f) $[\mathbf{L} \cdot \mathbf{S}, J^2]$. *Hint:* $\mathbf{L}$ and $\mathbf{S}$ satisfy the fundamental commutation relations for angular momentum (Equations 4.98, 4.99, and 4.134), but they commute with each other.

***Problem 6.15** Derive the fine structure formula (Equation 6.65) from the relativistic correction (Equation 6.56) and the spin-orbit coupling (Equation 6.64). *Hint:* Note that $j = l \pm 1/2$; treat the plus sign and the minus sign separately, and you'll find that you get the same final answer either way.

****Problem 6.16** The most prominent feature of the hydrogen spectrum in the visible region is the red Balmer line, coming from the transition $n = 3$ to $n = 2$. First of all, determine the wavelength and frequency of this line, according to the Bohr theory. Fine structure splits this line into several closely spaced lines; the question is: *How many*, and *what is their spacing*? *Hint:* First determine how many sublevels the $n = 2$ level splits into, and find E_{fs}^1 for each of these, in eV. Then do the same for $n = 3$. Draw an energy level diagram showing all possible transitions from $n = 3$ to $n = 2$. The energy released (in the form of a photon) is $(E_3 - E_2) + \Delta E$, the first part being common to all of them, and the ΔE (due to fine structure) varying from one transition to the next. Find ΔE (in eV) for each transition. Finally, convert to photon frequency, and determine the spacing between adjacent spectral lines (in Hz)—*not* the frequency interval between each line and the *unperturbed* line (which

is, of course, unobservable), but the frequency interval between each line and the *next* one. Your final answer should take the form, “The red Balmer line splits into (??) lines. In order of increasing frequency, they come from the transitions (1) $j = (??)$ to $j = (??)$, (2) $j = (??)$ to $j = (??)$, and so on. The frequency spacing between line (1) and line (2) is (???) Hz, the spacing between line (2) and line (3) is (???) Hz, and so on.”

Problem 6.17 The *exact* fine-structure formula for hydrogen (obtained from the Dirac equation without recourse to perturbation theory) is¹¹

$$E_{nj} = mc^2 \left\{ \left[1 + \left(\frac{\alpha}{n - (j + 1/2) + \sqrt{(j + 1/2)^2 - \alpha^2}} \right)^2 \right]^{-1/2} - 1 \right\}.$$

Expand to order α^4 (noting that $\alpha \ll 1$), and show that you recover Equation 6.66.

6.4 THE ZEEMAN EFFECT

When an atom is placed in a uniform external magnetic field $\mathbf{B}_{\text{ext}}$, the energy levels are shifted. This phenomenon is known as the **Zeeman effect**. For a single electron, the perturbation is

$$H'_Z = -(\boldsymbol{\mu}_l + \boldsymbol{\mu}_s) \cdot \mathbf{B}_{\text{ext}}, \quad [6.67]$$

where

$$\boldsymbol{\mu}_s = -\frac{e}{m} \mathbf{S} \quad [6.68]$$

is the magnetic dipole moment associated with electron spin, and

$$\boldsymbol{\mu}_l = -\frac{e}{2m} \mathbf{L} \quad [6.69]$$

is the dipole moment associated with orbital motion.¹² Thus

$$H'_Z = \frac{e}{2m} (\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}_{\text{ext}}. \quad [6.70]$$

The nature of the Zeeman splitting depends critically on the strength of the external field in comparison with the *internal* field (Equation 6.58) that gives rise to spin-orbit coupling. For if $B_{\text{ext}} \ll B_{\text{int}}$, then fine structure dominates, and H_Z can be treated as a small perturbation, whereas if $B_{\text{ext}} \gg B_{\text{int}}$, then the Zeeman

¹¹Bethe and Salpeter (footnote 7) page 83.

¹²The gyromagnetic ratio for *orbital* motion is just the classical value ($q/2m$)—it is only for *spin* that there is an “extra” factor of 2.

effect dominates, and fine structure becomes the perturbation. In the intermediate zone, where the two fields are comparable, we need the full machinery of degenerate perturbation theory, and it is necessary to diagonalize the relevant portion of the Hamiltonian “by hand”. In the following sections we shall explore each of these regimes briefly, for the case of hydrogen.

Problem 6.18 Use Equation 6.58 to estimate the internal field in hydrogen, and characterize quantitatively a “strong” and “weak” Zeeman field.

6.4.1 Weak-Field Zeeman Effect

If $B_{\text{ext}} \ll B_{\text{int}}$, fine structure dominates (Equation 6.66); the “good” quantum numbers are n , l , j , and m_j (but not m_l and m_s , because—in the presence of spin-orbit coupling— $\mathbf{L}$ and $\mathbf{S}$ are not separately conserved). In first-order perturbation theory, the Zeeman correction to the energy is

$$E_Z^1 = \langle n l j m_j | H'_Z | n l j m_j \rangle = \frac{e}{2m} \mathbf{B}_{\text{ext}} \cdot \langle \mathbf{L} + 2\mathbf{S} \rangle. \quad [6.71]$$

Now $\mathbf{L} + 2\mathbf{S} = \mathbf{J} + \mathbf{S}$; unfortunately, we do not immediately know the expectation value of $\mathbf{S}$. But we can figure it out as follows: The total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is constant (Figure 6.10); $\mathbf{L}$ and $\mathbf{S}$ precess rapidly about this fixed vector. In particular, the (time) *average* value of $\mathbf{S}$ is just its projection along $\mathbf{J}$:

$$\mathbf{S}_{\text{ave}} = \frac{(\mathbf{S} \cdot \mathbf{J})}{J^2} \mathbf{J}. \quad [6.72]$$

But $\mathbf{L} = \mathbf{J} - \mathbf{S}$, so $L^2 = J^2 + S^2 - 2\mathbf{J} \cdot \mathbf{S}$, and hence

$$\mathbf{S} \cdot \mathbf{J} = \frac{1}{2}(J^2 + S^2 - L^2) = \frac{\hbar^2}{2}[j(j+1) + s(s+1) - l(l+1)], \quad [6.73]$$

from which it follows that

$$\langle \mathbf{L} + 2\mathbf{S} \rangle = \left\langle \left(1 + \frac{\mathbf{S} \cdot \mathbf{J}}{J^2} \right) \mathbf{J} \right\rangle = \left[1 + \frac{j(j+1) - l(l+1) + 3/4}{2j(j+1)} \right] \langle \mathbf{J} \rangle. \quad [6.74]$$

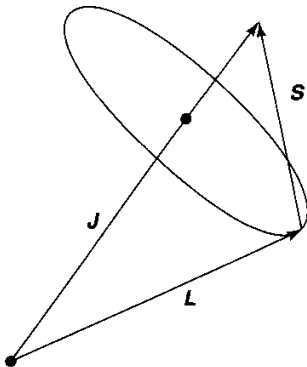


Figure 6.10: In the presence of spin-orbit coupling, $\mathbf{L}$ and $\mathbf{S}$ are not separately conserved; they precess about the fixed total angular momentum, $\mathbf{J}$.

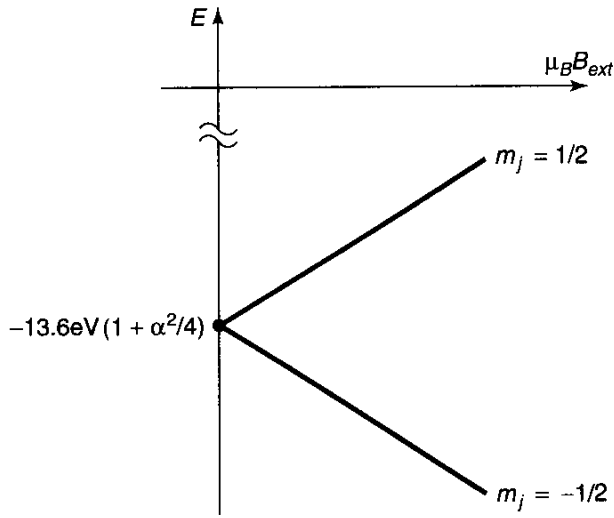


Figure 6.11: Weak-field Zeeman splitting of the ground state; the upper line ($m_j = 1/2$) has slope 1, the lower line ($m_j = -1/2$) has slope -1

The term in square brackets is known as the **Landé g-factor**, g_J .

We may as well choose the z -axis to lie along $\mathbf{B}_{\text{ext}}$; then

$$E_Z^1 = \mu_B g_J B_{\text{ext}} m_j, \quad [6.75]$$

where

$$\mu_B \equiv \frac{e\hbar}{2m} = 5.788 \times 10^{-5} \text{ eV/T} \quad [6.76]$$

is the so-called **Bohr magneton**. The *total* energy is the sum of the fine-structure part (Equation 6.66) and the Zeeman contribution (Equation 6.75). For example, the ground state ($n = 1$, $l = 0$, $j = 1/2$, and therefore $g_J = 2$) splits into two levels:

$$-13.6 \text{ eV}(1 + \alpha^2/4) \pm \mu_B B_{\text{ext}}, \quad [6.77]$$

with the plus sign for $m_j = 1/2$, and minus for $m_j = -1/2$. These energies are plotted (as functions of B_{ext}) in Figure 6.11.

***Problem 6.19** Consider the (eight) $n = 2$ states, $|2 \ l \ j \ m_j\rangle$. Find the energy of each state, under weak-field Zeeman splitting, and construct a diagram like Figure 6.11 to show how the energies evolve as B_{ext} increases. Label each line clearly, and indicate its slope.

6.4.2 Strong-Field Zeeman Effect

If $B_{\text{ext}} \gg B_{\text{int}}$, the Zeeman effect dominates¹³; with B_{ext} in the z direction, the “good” quantum numbers are now n , l , m_l , and m_s (but not j and m_j because—in the presence of the external torque—the total angular momentum is not conserved, whereas L_z and

¹³In this regime the Zeeman effect is also known as the **Paschen-Back effect**.

S_z are). The Zeeman Hamiltonian is

$$H'_Z = \frac{e}{2m} B_{\text{ext}}(L_z + 2S_z),$$

and the “unperturbed” energies are

$$E_{nm_l m_s} = -\frac{13.6 \text{ eV}}{n^2} + \mu_B B_{\text{ext}}(m_l + 2m_s). \quad [6.78]$$

In first-order perturbation theory, the fine-structure correction to these levels is

$$E_{\text{fs}}^1 = \langle n l m_l m_s | (H'_r + H'_{so}) | n l m_l m_s \rangle. \quad [6.79]$$

The relativistic contribution is the same as before (Equation 6.56); for the spin-orbit term (Equation 6.60) we need

$$\langle \mathbf{S} \cdot \mathbf{L} \rangle = \langle S_x \rangle \langle L_x \rangle + \langle S_y \rangle \langle L_y \rangle + \langle S_z \rangle \langle L_z \rangle = \hbar^2 m_l m_s \quad [6.80]$$

(note that $\langle S_x \rangle = \langle S_y \rangle = \langle L_x \rangle = \langle L_y \rangle = 0$ for eigenstates of S_z and L_z). Putting all this together (Problem 6.20), we conclude that

$$E_{\text{fs}}^1 = \frac{13.6 \text{ eV}}{n^3} \alpha^2 \left\{ \frac{3}{4n} - \left[\frac{l(l+1) - m_l m_s}{l(l+1/2)(l+1)} \right] \right\}. \quad [6.81]$$

(The term in square brackets is indeterminate for $l = 0$; its correct value in this case is 1—see Problem 6.22.) The *total* energy is the sum of the Zeeman part (Equation 6.78) and the fine-structure contribution (Equation 6.81).

Problem 6.20 Starting with Equation 6.79 and using Equations 6.56, 6.60, 6.63, and 6.80, derive Equation 6.81.

****Problem 6.21** Consider the eight $n = 2$ states, $|2 l m_l m_s\rangle$. Find the energy of each state, under strong-field Zeeman splitting. (Express your answers as the sum of three terms, as in Equation 6.77: the Bohr energy; the fine structure, proportional to α^2 ; and the Zeeman contribution, proportional to $\mu_B B_{\text{ext}}$.) If you ignore fine structure altogether, how many distinct levels are there, and what are their degeneracies?

Problem 6.22 If $l = 0$, then $j = s$, $m_j = m_s$, and the “good” states are the same ($|n m_s\rangle$) for weak *and* strong fields. Determine E_Z^1 (from Equation 6.71) and the fine structure energies (Equation 6.66), and write down the general result for the $l = 0$ Zeeman effect—*regardless* of the strength of the field. Show that the strong-field formula (Equation 6.81) reproduces this result, provided that we interpret the indeterminate term in square brackets as 1.

6.4.3 Intermediate-Field Zeeman Effect

In the intermediate regime, neither H'_Z nor H'_{fs} dominates, and we must treat the two on an equal footing, as perturbations to the Bohr Hamiltonian (Equation 6.41):

$$H' = H'_Z + H'_{fs}. \quad [6.82]$$

I'll confine my attention here to the case $n = 2$ and choose as the basis for degenerate perturbation theory the states characterized by l , j , and m_j .¹⁴ Using the Clebsch-Gordan coefficients (Problem 4.45 or Table 4.7) to express $|j m_j\rangle$ as a linear combination of $|l m_l\rangle|s m_s\rangle$, we have

$$l = 0 \begin{cases} \psi_1 \equiv |\frac{1}{2} \frac{1}{2}\rangle = |0 0\rangle|\frac{1}{2} \frac{1}{2}\rangle, \\ \psi_2 \equiv |\frac{1}{2} \frac{-1}{2}\rangle = |0 0\rangle|\frac{1}{2} \frac{-1}{2}\rangle, \end{cases}$$

$$l = 1 \begin{cases} \psi_3 \equiv |\frac{3}{2} \frac{3}{2}\rangle = |1 1\rangle|\frac{1}{2} \frac{1}{2}\rangle, \\ \psi_4 \equiv |\frac{3}{2} \frac{-3}{2}\rangle = |1 -1\rangle|\frac{1}{2} \frac{-1}{2}\rangle, \\ \psi_5 \equiv |\frac{3}{2} \frac{1}{2}\rangle = \sqrt{2/3}|1 0\rangle|\frac{1}{2} \frac{1}{2}\rangle + \sqrt{1/3}|1 1\rangle|\frac{1}{2} \frac{-1}{2}\rangle, \\ \psi_6 \equiv |\frac{1}{2} \frac{1}{2}\rangle = -\sqrt{1/3}|1 0\rangle|\frac{1}{2} \frac{1}{2}\rangle + \sqrt{2/3}|1 1\rangle|\frac{1}{2} \frac{-1}{2}\rangle, \\ \psi_7 \equiv |\frac{3}{2} \frac{-1}{2}\rangle = \sqrt{1/3}|1 -1\rangle|\frac{1}{2} \frac{1}{2}\rangle + \sqrt{2/3}|1 0\rangle|\frac{1}{2} \frac{-1}{2}\rangle, \\ \psi_8 \equiv |\frac{1}{2} \frac{-1}{2}\rangle = -\sqrt{2/3}|1 -1\rangle|\frac{1}{2} \frac{1}{2}\rangle + \sqrt{1/3}|1 0\rangle|\frac{1}{2} \frac{-1}{2}\rangle. \end{cases}$$

In this basis the nonzero matrix elements of H'_{fs} are all on the diagonal, and given by Equation 6.65; H'_Z has four off-diagonal elements, and the complete matrix $-W$ is (see Problem 6.23)

$$\begin{pmatrix} 5\gamma - \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5\gamma + \beta & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma - 2\beta & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \gamma + 2\beta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \gamma - \frac{2}{3}\beta & \frac{\sqrt{2}}{3}\beta & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{3}\beta & 5\gamma - \frac{1}{3}\beta & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \gamma + \frac{2}{3}\beta & \frac{\sqrt{2}}{3}\beta \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{3}\beta & 5\gamma + \frac{1}{3}\beta \end{pmatrix}$$

where

$$\gamma \equiv (\alpha/8)^2 13.6 \text{ eV} \quad \text{and} \quad \beta \equiv \mu_B B_{\text{ext}}.$$

The first four eigenvalues are displayed along the diagonal; it remains only to find the eigenvalues of the two 2×2 blocks. The characteristic equation for the first is

$$\lambda^2 + \lambda(6\gamma - \beta) + (5\gamma^2 - \frac{11}{3}\gamma\beta) = 0,$$

¹⁴You can use l, m_l, m_s states if you prefer—this makes the matrix elements of H'_Z easier but those of H'_{fs} more difficult; the W -matrix will be more complicated, but its eigenvalues (which are independent of basis) are the same either way.

Table 6.2: Energy levels for the $n = 2$ states of hydrogen, with fine structure and Zeeman splitting.

ϵ_1	$=$	$E_2 - 5\gamma + \beta$
ϵ_2	$=$	$E_2 - 5\gamma - \beta$
ϵ_3	$=$	$E_2 - \gamma + 2\beta$
ϵ_4	$=$	$E_2 - \gamma - 2\beta$
ϵ_5	$=$	$E_2 - 3\gamma + \beta/2 + \sqrt{4\gamma^2 + (2/3)\gamma\beta + \beta^2/4}$
ϵ_6	$=$	$E_2 - 3\gamma + \beta/2 - \sqrt{4\gamma^2 + (2/3)\gamma\beta + \beta^2/4}$
ϵ_7	$=$	$E_2 - 3\gamma - \beta/2 + \sqrt{4\gamma^2 - (2/3)\gamma\beta + \beta^2/4}$
ϵ_8	$=$	$E_2 - 3\gamma - \beta/2 - \sqrt{4\gamma^2 - (2/3)\gamma\beta + \beta^2/4}$

and the quadratic formula gives the eigenvalues:

$$\lambda_{\pm} = -3\gamma + (\beta/2) \pm \sqrt{4\gamma^2 + (2/3)\gamma\beta + (\beta^2/4)}. \quad [6.83]$$

The eigenvalues of the second block are the same, but with the sign of β reversed. The eight energies are listed in Table 6.2, and plotted against B_{ext} in Figure 6.12. In the zero-field limit ($\beta = 0$) they reduce to the fine-structure values; for weak fields ($\beta \ll \gamma$) they reproduce what you got in Problem 6.19; for strong fields ($\beta \gg \gamma$) we recover the results of Problem 6.21 (note the convergence to five distinct energy levels, at very high fields, as predicted in Problem 6.21).

Problem 6.23 Work out the matrix elements of H'_Z and H'_{fs} , and construct the W -matrix given in the text, for $n = 2$.

*****Problem 6.24** Analyze the Zeeman effect for the $n = 3$ states of hydrogen in the weak, strong, and intermediate field regimes. Construct a table of energies (analogous

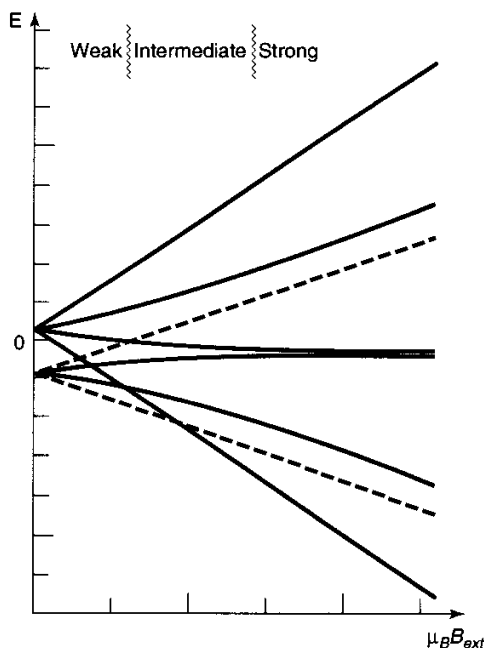


Figure 6.12: Zeeman splitting of the $n = 2$ states of hydrogen in the weak, intermediate, and strong field regimes.

to Table 6.2), plot them as functions of the external field (as in Figure 6.12), and check that the intermediate-field results reduce properly in the two limiting cases.

6.5 HYPERFINE SPLITTING

The proton itself constitutes a magnetic dipole, though its dipole moment is much smaller than the electron's because of the mass in the denominator (Equation 6.59):

$$\boldsymbol{\mu}_p = \frac{ge}{2m_p} \mathbf{S}_p, \quad \boldsymbol{\mu}_e = -\frac{e}{m_e} \mathbf{S}_e. \quad [6.84]$$

(The proton is a composite structure, made up of three quarks, and its gyromagnetic ratio is not as simple as the electron's—hence the g -factor,¹⁵ whose measured value is 5.59 as opposed to 2.00 for the electron.) According to classical electrodynamics, a dipole $\boldsymbol{\mu}$ sets up a magnetic field¹⁶

$$\mathbf{B} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \boldsymbol{\mu}] + \frac{2\mu_0}{3} \boldsymbol{\mu} \delta^3(\mathbf{r}). \quad [6.85]$$

So the Hamiltonian (Equation 6.57) of the electron, in the magnetic field due to the proton's magnetic dipole moment, is

$$H'_{\text{hf}} = \frac{\mu_0 ge^2}{8\pi m_p m_e} \frac{[3(\mathbf{S}_p \cdot \hat{\mathbf{r}})(\mathbf{S}_e \cdot \hat{\mathbf{r}}) - \mathbf{S}_p \cdot \mathbf{S}_e]}{r^3} + \frac{\mu_0 ge^2}{3m_p m_e} \mathbf{S}_p \cdot \mathbf{S}_e \delta^3(\mathbf{r}). \quad [6.86]$$

According to perturbation theory, the first-order correction to the energy (Equation 6.9) is the expectation value of the perturbing Hamiltonian:

$$\begin{aligned} E_{\text{hf}}^1 &= \frac{\mu_0 ge^2}{8\pi m_p m_e} \left\langle \frac{3(\mathbf{S}_p \cdot \hat{\mathbf{r}})(\mathbf{S}_e \cdot \hat{\mathbf{r}}) - \mathbf{S}_p \cdot \mathbf{S}_e}{r^3} \right\rangle \\ &\quad + \frac{\mu_0 ge^2}{3m_p m_e} \langle \mathbf{S}_p \cdot \mathbf{S}_e \rangle |\psi(0)|^2. \end{aligned} \quad [6.87]$$

In the ground state (or any other state for which $l = 0$) the wave function is spherically symmetrical, and the first expectation value vanishes (see Problem 6.25). Meanwhile, from Equation 4.80 we find that $|\psi_{100}(0)|^2 = 1/(\pi a^3)$, so

$$E_{\text{hf}}^1 = \frac{\mu_0 ge^2}{3\pi m_p m_e a^3} \langle \mathbf{S}_p \cdot \mathbf{S}_e \rangle, \quad [6.88]$$

¹⁵The Landé g -factor, in Equation 6.74, plays a similar role in the proportionality between the electron's *total* magnetic moment ($\boldsymbol{\mu}_l + \boldsymbol{\mu}_s$) and its total angular momentum $\mathbf{J}$.

¹⁶If you are unfamiliar with the delta function term in Equation 6.85, you can derive it by treating the dipole as a spinning charged spherical shell, in the limit as the radius goes to zero and the charge goes to infinity (with $\boldsymbol{\mu}$ held constant). See D. J. Griffiths, *Am. J. Phys.* **50**, 698 (1982).

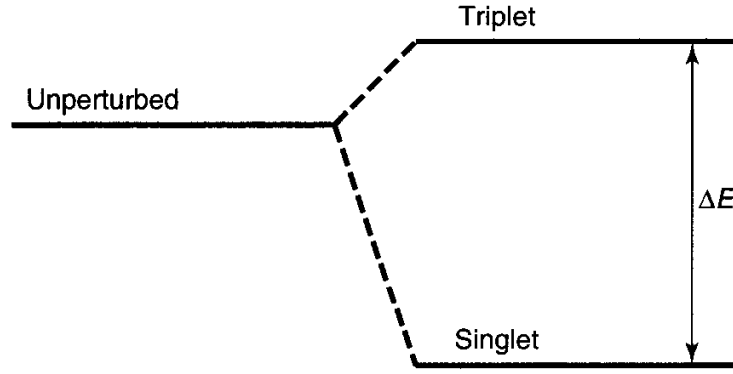


Figure 6.13: Hyperfine splitting in the ground state of hydrogen.

in the ground state. This is called **spin-spin coupling** because it involves the dot product of two spins (contrast spin-orbit coupling, which involves $\mathbf{S} \cdot \mathbf{L}$).

In the presence of spin-spin coupling, the individual spin angular momenta are no longer conserved; the “good” states are eigenvectors of the *total* spin,

$$\mathbf{S} \equiv \mathbf{S}_e + \mathbf{S}_p. \quad [6.89]$$

As before, we square this out to get

$$\mathbf{S}_p \cdot \mathbf{S}_e = \frac{1}{2}(S^2 - S_e^2 - S_p^2). \quad [6.90]$$

But the electron and proton both have spin 1/2, so $S_e^2 = S_p^2 = (3/4)\hbar^2$. In the triplet state (spins “parallel”) the total spin is 1, and hence $S^2 = 2\hbar^2$; in the singlet state the total spin is 0, and $S^2 = 0$. Thus

$$E_{\text{hf}}^1 = \frac{4g\hbar^4}{3m_p m_e^2 c^2 a^4} \begin{cases} +1/4, & (\text{triplet}); \\ -3/4, & (\text{singlet}). \end{cases} \quad [6.91]$$

Spin-spin coupling breaks the spin degeneracy of the ground state, lifting the triplet configuration and depressing the singlet (see Figure 6.13).

The energy gap is evidently

$$\Delta E = \frac{4g\hbar^4}{3m_p m_e^2 c^2 a^4} = 5.88 \times 10^{-6} \text{ eV}. \quad [6.92]$$

The frequency of the photon emitted in a transition from the triplet to the singlet state is

$$\nu = \frac{\Delta E}{h} = 1420 \text{ MHz}, \quad [6.93]$$

and the corresponding wavelength is $c/\nu = 21 \text{ cm}$, which falls in the microwave region. This famous “21-centimeter line” is among the most pervasive and ubiquitous forms of radiation in the universe.

Problem 6.25 Let $\mathbf{a}$ and $\mathbf{b}$ be two constant vectors. Show that

$$\int (\mathbf{a} \cdot \hat{\mathbf{r}})(\mathbf{b} \cdot \hat{\mathbf{r}}) \sin \theta \, d\theta \, d\phi = \frac{4\pi}{3} (\mathbf{a} \cdot \mathbf{b}). \quad [6.94]$$

The integration is over the usual range: $0 < \theta < \pi$, $0 < \phi < 2\pi$. Use this result to demonstrate that

$$\left\langle \frac{3(\mathbf{S}_p \cdot \hat{\mathbf{r}})(\mathbf{S}_e \cdot \hat{\mathbf{r}}) - \mathbf{S}_p \cdot \mathbf{S}_e}{r^3} \right\rangle = 0,$$

for states with $l = 0$. *Hint:* $\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}$.

Problem 6.26 By appropriate modification of the hydrogen formula, determine the hyperfine splitting in the ground state of (a) muonic hydrogen (in which a muon—same charge and g -factor as the electron, but 207 times the mass—substitutes for the electron), (b) positronium (in which a positron—same mass and g -factor as the electron, but opposite charge—substitutes for the proton), and (c) muonium (in which an antimuon—same mass and g -factor as a muon, but opposite charge—substitutes for the proton). *Hint:* Don't forget to use the reduced mass (Problem 5.1) in calculating the “Bohr radius” of these exotic “atoms.”. Incidentally, the answer you get for positronium (4.85×10^{-4} eV) is quite far from the experimental value (8.41×10^{-4} eV); the large discrepancy is due to pair annihilation ($e^+ + e^- \rightarrow \gamma + \gamma$), which contributes an extra $(3/4)\Delta E$ and does not occur (of course) in ordinary hydrogen, muonic hydrogen, or muonium. See Griffiths (footnote 16) for further details.

FURTHER PROBLEMS FOR CHAPTER 6

****Problem 6.27** Suppose the Hamiltonian H , for a particular quantum system, is a function of some parameter λ ; let $E_n(\lambda)$ and $\psi_n(\lambda)$ be the eigenvalues and eigenfunctions of $H(\lambda)$. The **Feynman-Hellmann theorem** states that

$$\frac{\partial E_n}{\partial \lambda} = \langle \psi_n | \frac{\partial H}{\partial \lambda} | \psi_n \rangle \quad [6.95]$$

(assuming either that E_n is nondegenerate, or—if degenerate—that the ψ_n 's are the “good” linear combinations of the degenerate eigenfunctions).

- (a) Prove the Feynman-Hellmann theorem. *Hint:* Use Equation 6.9.
- (b) Apply it to the one-dimensional harmonic oscillator, (i) using $\lambda = \omega$ (this yields a formula for the expectation value of V), (ii) using $\lambda = \hbar$ (this yields $\langle T \rangle$), and (iii) using $\lambda = m$ (this yields a relation between $\langle T \rangle$ and $\langle V \rangle$). Compare your answers to Problem 2.37 and the virial theorem predictions (Problem 3.53).

****Problem 6.28** The Feynman-Hellmann theorem (Problem 6.27) can be used to determine the expectation values of $1/r$ and $1/r^2$ for hydrogen.¹⁷ The effective Hamiltonian for the radial wave functions is (Equation 4.53)

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r},$$

and the eigenvalues (expressed in terms of l)¹⁸ are (Equation 4.70)

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2(j_{\max} + l + 1)^2}.$$

(a) Use $\lambda = e$ in the Feynman-Hellmann theorem to obtain $\langle 1/r \rangle$. Check your result against Equation 6.54.

(b) Use $\lambda = l$ to obtain $\langle 1/r^2 \rangle$. Check your answer with Equation 6.55.

*****Problem 6.29** Prove **Kramers' relation**:

$$\frac{s+1}{n^2} \langle r^s \rangle - (2s+1)a \langle r^{s-1} \rangle + \frac{s}{4} [(2l+1)^2 - s^2] a^2 \langle r^{s-2} \rangle = 0, \quad [6.96]$$

which relates the expectation values of r to three different powers (s , $s-1$, and $s-2$), for an electron in the state ψ_{nlm} of hydrogen. *Hint*: Rewrite the radial equation (Equation 4.53) in the form

$$u'' = \left[\frac{l(l+1)}{r^2} - \frac{2}{ar} + \frac{1}{n^2 a^2} \right] u,$$

and use it to express $\int (ur^s u'') dr$ in terms of $\langle r^s \rangle$, $\langle r^{s-1} \rangle$, and $\langle r^{s-2} \rangle$. Then use integration by parts to reduce the second derivative. Show that $\int (ur^s u') dr = -(s/2) \langle r^{s-1} \rangle$, and $\int (u' r^s u') dr = -[2/(s+1)] \int (u'' r^{s+1} u') dr$. Take it from there.

Problem 6.30

(a) Plug $s = 0$, $s = 1$, $s = 2$, and $s = 3$ into Kramers' relation (Equation 6.96) to obtain formulas for $\langle r^{-1} \rangle$, $\langle r \rangle$, $\langle r^2 \rangle$, and $\langle r^3 \rangle$. Note that you could continue indefinitely, to find *any* positive power.

(b) In the *other* direction, however, you hit a snag. Put in $s = -1$, and show that all you get is a relation between $\langle r^{-2} \rangle$ and $\langle r^{-3} \rangle$.

¹⁷C. Sánchez del Río, *Am. J. Phys.*, **50**, 556 (1982); H. S. Valk, *Am. J. Phys.*, **54**, 921 (1986).

¹⁸In part (b) we treat l as a continuous variable; n becomes a function of l , according to Equation 4.67, because $j_{\max}$, which must be an integer, is fixed. To avoid confusion, I have eliminated n , to reveal the dependence on l explicitly.

- (c) But if you can get $\langle r^{-2} \rangle$ by some *other* means, you can apply the Kramers' relation to obtain the rest of the negative powers. Use Equation 6.55 (which is derived in Problem 6.28) to determine $\langle r^{-3} \rangle$, and check your answer against Equation 6.63.

*****Problem 6.31** When an atom is placed in a uniform external electric field $\mathbf{E}_{\text{ext}}$, the energy levels are shifted—a phenomenon known as the **Stark effect**. In this problem we analyze the Stark effect for the $n = 1$ and $n = 2$ states of hydrogen. Let the field point in the z direction, so the potential energy of the electron is

$$H'_S = -eE_{\text{ext}}z = -eE_{\text{ext}}r \cos \theta.$$

Treat this as a perturbation on the Bohr Hamiltonian (Equation 6.41); spin is irrelevant to this problem, so ignore it.

- (a) Show that the ground-state energy is not affected by this perturbation, in first order.
- (b) The first excited state is fourfold degenerate: $\psi_{200}, \psi_{211}, \psi_{210}, \psi_{21-1}$. Using degenerate perturbation theory, determine the first-order corrections to the energy. Into how many levels does E_2 split?
- (c) What are the “good” wave functions for part (b)? Find the expectation value of the electric dipole moment ($\mathbf{p}_e = -e\mathbf{r}$), in each of these “good” states. Notice that the results are independent of the applied field—evidently hydrogen in its first excited state can carry a *permanent* electric dipole moment.

Hint: There are a lot of integrals in this problem, but almost all of them are zero. So study each one carefully before you do any calculations: If the ϕ integral vanishes, there's not much point in doing the r and θ integrals! *Partial answer:* $W_{13} = W_{31} = 3eaE_{\text{ext}}$; all other elements are zero.

*****Problem 6.32** Consider the Stark effect (Problem 6.31) for the $n = 3$ states of hydrogen. There are initially nine degenerate states, ψ_{3lm} (neglecting spin, of course), and we turn on an electric field in the z direction.

- (a) Construct the 9×9 matrix representing the perturbing Hamiltonian. *Partial answer:* $\langle 3\ 0\ 0 | z | 3\ 1\ 0 \rangle = -3\sqrt{6}a$, $\langle 3\ 1\ 0 | z | 3\ 2\ 0 \rangle = -3\sqrt{3}a$, $\langle 3\ 1\ \pm 1 | z | 3\ 2\ \pm 1 \rangle = -(9/2)a$.
- (b) Find the eigenvalues and their degeneracies.

Problem 6.33 Calculate the wavelength, in centimeters, of the photon emitted under a hyperfine transition in the ground state ($n = 1$) of **deuterium**. Deuterium is “heavy” hydrogen, with an extra neutron in the nucleus. The proton and neutron bind together to form a **deuteron**, with spin 1 and magnetic moment

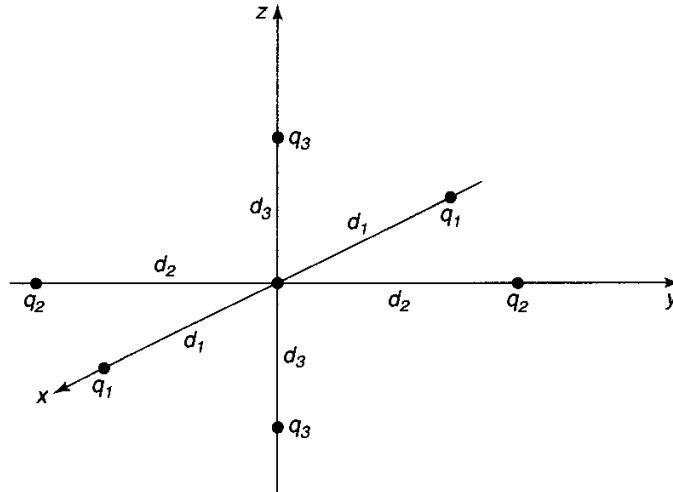


Figure 6.14: Hydrogen atom surrounded by six point charges (crude model for a crystal lattice); Problem 6.34.

$$\mu_d = \frac{g_d e}{2m_d} S_d;$$

the deuteron g -factor is 1.71.

*****Problem 6.34** In a crystal, the electric field of neighboring ions perturbs the energy levels of an atom. As a crude model, imagine that a hydrogen atom is surrounded by three pairs of point charges, as shown in Figure 6.14. (Spin is irrelevant to this problem, so ignore it.)

(a) Assuming that $r \ll d_1$, $r \ll d_2$, and $r \ll d_3$, show that

$$H' = V_0 + 3(\beta_1 x^2 + \beta_2 y^2 + \beta_3 z^2) - (\beta_1 + \beta_2 + \beta_3)r^2,$$

where

$$\beta_i \equiv -\frac{e}{4\pi\epsilon_0} \frac{q_i}{d_i^3}, \quad \text{and } V_0 = 2(\beta_1 d_1^2 + \beta_2 d_2^2 + \beta_3 d_3^2).$$

(b) Find the lowest-order correction to the ground-state energy.

(c) Calculate the first-order corrections to the energy of the first excited states ($n = 2$). Into how many levels does this fourfold degenerate system split, (i) in the case of **cubic symmetry**, $\beta_1 = \beta_2 = \beta_3$; (ii) in the case of **orthorhombic symmetry**, $\beta_1 = \beta_2 \neq \beta_3$; (iii) in the general case of **tetragonal symmetry** (all three different)?

CHAPTER 7

THE VARIATIONAL PRINCIPLE

7.1 THEORY

Suppose you want to calculate the ground-state energy E_g for a system described by the Hamiltonian H , but you are unable to solve the (time-independent) Schrödinger equation. Pick *any normalized function ψ whatsoever*.

Theorem:

$$E_g \leq \langle \psi | H | \psi \rangle \equiv \langle H \rangle. \quad [7.1]$$

That is, the expectation value of H in the (presumably incorrect) state ψ is certain to *overestimate* the ground-state energy. Of course, if ψ just happens to be one of the *excited* states, then *obviously* $\langle H \rangle$ exceeds E_g ; but the theorem says that the same holds for any ψ whatsoever.

Proof: Since the (unknown) eigenfunctions of H form a complete set, we can express ψ as a linear combination of them¹:

$$\psi = \sum_n c_n \psi_n, \quad \text{with } H\psi_n = E_n \psi_n.$$

¹If the Hamiltonian admits scattering states, as well as bound states, then we'll need an integral as well as a sum, but the argument is unchanged.

Since ψ is normalized,

$$1 = \langle \psi | \psi \rangle = \left\langle \sum_m c_m \psi_m \middle| \sum_n c_n \psi_n \right\rangle = \sum_m \sum_n c_m^* c_n \langle \psi_m | \psi_n \rangle = \sum_n |c_n|^2$$

(assuming the eigenfunctions have been orthonormalized: $\langle \psi_m | \psi_n \rangle = \delta_{mn}$). Meanwhile,

$$\langle H \rangle = \left\langle \sum_m c_m \psi_m \middle| H \sum_n c_n \psi_n \right\rangle = \sum_m \sum_n c_m^* E_n c_n \langle \psi_m | \psi_n \rangle = \sum_n E_n |c_n|^2.$$

But the ground-state energy is, by definition, the *smallest* eigenvalue, so $E_g \leq E_n$, and hence

$$\langle H \rangle \geq E_g \sum_n |c_n|^2 = E_g.$$

QED

Example 1. Suppose we want to find the ground-state energy for the one-dimensional harmonic oscillator:

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2.$$

Of course, we already know the *exact* answer, in this case (Equation 2.49): $E_g = (1/2)\hbar\omega$; but this makes it a good test of the method. We might pick as our “trial” wave function the gaussian,

$$\psi(x) = A e^{-bx^2}, \quad [7.2]$$

where b is a constant and A is determined by normalization:

$$1 = |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} dx = |A|^2 \sqrt{\frac{\pi}{2b}} \Rightarrow A = \left(\frac{2b}{\pi} \right)^{1/4}. \quad [7.3]$$

Now

$$\langle H \rangle = \langle T \rangle + \langle V \rangle, \quad [7.4]$$

where, in this case,

$$\langle T \rangle = -\frac{\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} e^{-bx^2} \frac{d^2}{dx^2} (e^{-bx^2}) dx = \frac{\hbar^2 b}{2m}, \quad [7.5]$$

and

$$\langle V \rangle = \frac{1}{2} m \omega^2 |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} x^2 dx = \frac{m \omega^2}{8b},$$

so

$$\langle H \rangle = \frac{\hbar^2 b}{2m} + \frac{m \omega^2}{8b}. \quad [7.6]$$

According to the theorem, this exceeds E_g for any b ; to get the *tightest* bound let's *minimize* $\langle H \rangle$ with respect to b :

$$\frac{d}{db} \langle H \rangle = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8b^2} = 0 \Rightarrow b = \frac{m\omega}{2\hbar}.$$

Putting this back into $\langle H \rangle$, we find

$$\langle H \rangle_{\min} = \frac{1}{2} \hbar \omega. \quad [7.7]$$

In this case we hit the ground-state energy right on the nose—because (obviously) I “just happened” to pick a trial function with precisely the form of the *actual* ground state (Equation 2.48). But the gaussian is very easy to work with, so it's a popular trial function even when it bears little resemblance to the true ground state.

Example 2. Suppose we're looking for the ground state energy of the delta-function potential:

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - \alpha \delta(x).$$

Again, we already know the exact answer (Equation 2.109): $E_g = -m\alpha^2/2\hbar^2$. As before, we'll use a gaussian trial function (Equation 7.2). We've already determined the normalization and calculated $\langle T \rangle$; all we need is

$$\langle V \rangle = -\alpha |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} \delta(x) dx = -\alpha \sqrt{\frac{2b}{\pi}}.$$

Evidently,

$$\langle H \rangle = \frac{\hbar^2 b}{2m} - \alpha \sqrt{\frac{2b}{\pi}}, \quad [7.8]$$

and we know that this exceeds E_g for all b . Minimizing it,

$$\frac{d}{db} \langle H \rangle = \frac{\hbar^2}{2m} - \frac{\alpha}{\sqrt{2\pi b}} = 0 \Rightarrow b = \frac{2m^2 \alpha^2}{\pi \hbar^4}.$$

So

$$\langle H \rangle_{\min} = -\frac{m\alpha^2}{\pi \hbar^2}, \quad [7.9]$$

which is indeed somewhat higher than E_g , since $\pi > 2$.

I said you can use *any* (normalized) trial function ψ whatsoever, and this is true in a sense. However, for *discontinuous* functions it takes some fancy footwork to assign a sensible interpretation to the second derivative (which you need, in order to calculate $\langle T \rangle$). Continuous functions with kinks in them are fair game, however: the next example shows how to handle them.

Example 3. Find an upper bound on the ground-state energy of the one-dimensional infinite square well (Equation 2.15), using the “triangular” trial wave function (Figure 7.1)²:

$$\psi(x) = \begin{cases} Ax, & \text{if } 0 \leq x \leq a/2, \\ A(a-x), & \text{if } a/2 \leq x \leq a, \\ 0, & \text{otherwise,} \end{cases} \quad [7.10]$$

where A is determined by normalization:

$$1 = |A|^2 \left[\int_0^{a/2} x^2 dx + \int_{a/2}^a (a-x)^2 dx \right] = |A|^2 \frac{a^3}{12} \Rightarrow A = \frac{2}{a} \sqrt{\frac{3}{a}}. \quad [7.11]$$

In this case

$$\frac{d\psi}{dx} = \begin{cases} A, & \text{if } 0 \leq x \leq a/2, \\ -A, & \text{if } a/2 \leq x \leq a, \\ 0, & \text{otherwise,} \end{cases}$$

as indicated in Figure 7.2. Now, the derivative of a step function is a delta function (see Problem 2.24b):

$$\frac{d^2\psi}{dx^2} = A\delta(x) - 2A\delta(x-a/2) + A\delta(x-a), \quad [7.12]$$

and hence

$$\begin{aligned} \langle H \rangle &= -\frac{\hbar^2 A}{2m} \int [\delta(x) - 2\delta(x-a/2) + \delta(x-a)] \psi(x) dx \\ &= -\frac{\hbar^2 A}{2m} [\psi(0) - 2\psi(a/2) + \psi(a)] = \frac{\hbar^2 A^2 a}{2m} = \frac{12\hbar^2}{2ma^2}. \end{aligned} \quad [7.13]$$

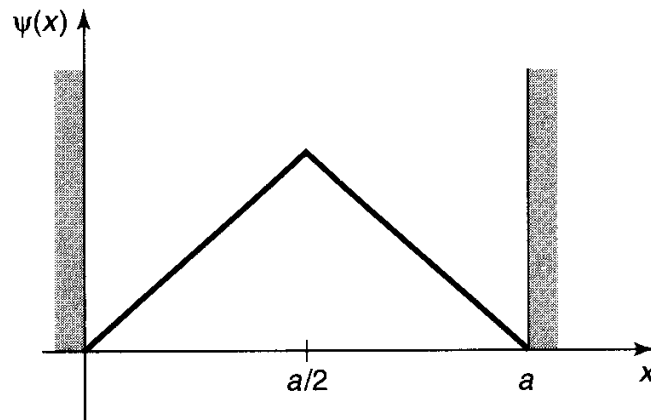


Figure 7.1: Triangular trial wave function for the infinite square well (Equation 7.10).

²There is no point in trying a function (such as the gaussian) that extends outside the well, because you’ll get $\langle V \rangle = \infty$, and Equation 7.1 tells you nothing.

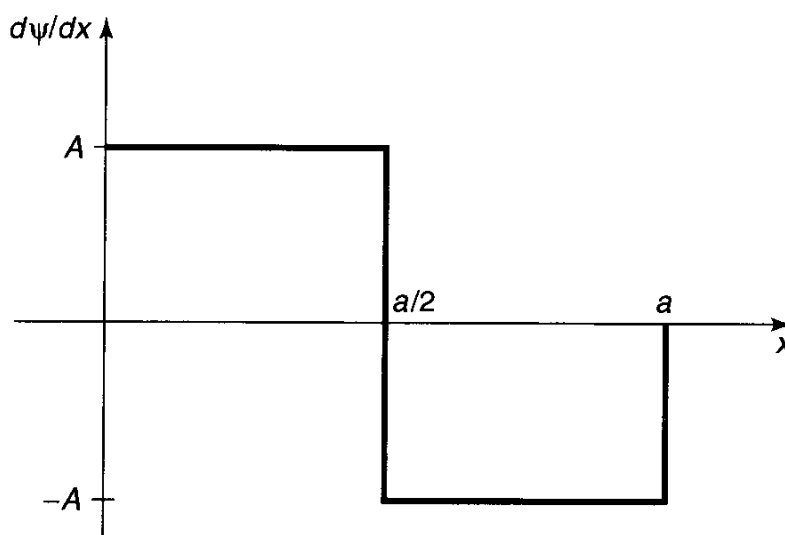


Figure 7.2: Derivative of the wave function in Figure 7.1.

The exact ground state is $E_g = \pi^2 \hbar^2 / 2ma^2$ (Equation 2.23), so the theorem works ($12 > \pi^2$).

The **variational principle** (as Equation 7.1 is called) is extremely powerful, and embarrassingly easy to use. What a chemist does, to find the ground-state energy of some complicated molecule, is write down a trial wave function with a large number of adjustable parameters, calculate $\langle H \rangle$, and tweak the parameters to get the lowest possible value. Even if ψ has no relation to the true wave function, one often gets miraculously accurate values for E_g . Naturally, if you have some way of guessing a *realistic* ψ , so much the better. The only *trouble* with the method is that you never know for sure how close you are to the target—all you can be *certain* of is that you've got an *upper bound*. Moreover, the technique applies only to the ground state (see, however, Problem 7.4).

***Problem 7.1** Use the gaussian trial function (Equation 7.2) to obtain the lowest upper bound you can on the ground-state energy of (a) the linear potential: $V(x) = \alpha|x|$; (b) the quartic potential: $V(x) = \alpha x^4$.

****Problem 7.2** Find the best bound on E_g for the one-dimensional harmonic oscillator using a trial wave function of the form

$$\psi(x) = \frac{A}{x^2 + b^2},$$

where A is determined by normalization and b is an adjustable parameter.

Problem 7.3 Find the best bound on E_g for the delta-function potential $-\alpha\delta(x - a/2)$, using the triangle trial function (Equation 7.10). (This time a is an adjustable parameter.)

Problem 7.4

- (a) Prove the following **corollary** to the variational principle: If $\langle \psi | \psi_g \rangle = 0$, then $\langle H \rangle \geq E_f$, where E_f is the energy of the first excited state.

Thus, if we can find a trial function that is orthogonal to the exact ground state, we can get an upper bound on the *first excited state*. In general, it's difficult to be sure that ψ is orthogonal to ψ_g , since (presumably) we don't *know* the latter. However, if the potential $V(x)$ is an *even* function of x , then the ground state is likewise even, and hence any *odd* trial function will automatically meet the condition for the corollary.

- (b) Find the best bound on the first excited state of the one-dimensional harmonic oscillator using the trial function

$$\psi(x) = A x e^{-bx^2}.$$

Problem 7.5

- (a) Use the variational principle to prove that first-order nondegenerate perturbation theory always *overestimates* (or at any rate never *underestimates*) the ground-state energy.
- (b) In view of (a), you would expect that the *second-order* correction to the ground state is always negative. Confirm that this is indeed the case, by examining Equation 6.14.

7.2 THE GROUND STATE OF HELIUM

The helium atom (Figure 7.3) consists of two electrons in orbit around a nucleus containing two protons (also some neutrons, which are irrelevant to our purpose). The Hamiltonian for this system (ignoring fine structure and smaller corrections) is

$$H = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\epsilon_0} \left(\frac{2}{r_1} + \frac{2}{r_2} - \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right). \quad [7.14]$$

Our problem is to calculate the ground-state energy, E_g —the amount of energy it would take to strip off the two electrons. (Given E_g it is easy to figure out the “ionization energy” required to remove a *single* electron—see Problem 7.6.) E_g has been measured very accurately in the laboratory:

$$E_g = -78.975 \text{ eV} \quad (\text{experimental}). \quad [7.15]$$

This is the number we would like to reproduce theoretically.

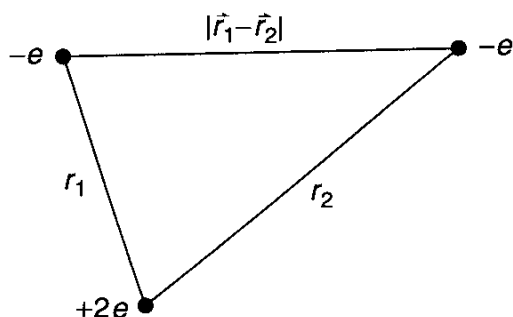


Figure 7.3: The helium atom.

It is curious that such a simple and important problem has no known exact solution.³ The trouble comes from the electron-electron repulsion,

$$V_{ee} = \frac{e^2}{4\pi\epsilon_0} \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}. \quad [7.16]$$

If we ignore this term altogether, H splits into two independent hydrogen Hamiltonians (only with a nuclear charge of $2e$, instead of e); the exact solution is just the product of hydrogenic wave functions:

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) \equiv \psi_{100}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2) = \frac{8}{\pi a^3} e^{-2(r_1+r_2)/a}, \quad [7.17]$$

and the energy is $8E_1 = -109$ eV (Eq. [5.31]).⁴ This is a long way from -79 eV, but it's a start.

To get a better approximation for E_g , we'll apply the variational principle, using ψ_0 as the trial wave function. This is a particularly convenient choice because it's an eigenfunction of *most* of the Hamiltonian:

$$H\psi_0 = (8E_1 + V_{ee})\psi_0. \quad [7.18]$$

Thus

$$\langle H \rangle = 8E_1 + \langle V_{ee} \rangle, \quad [7.19]$$

where⁵

$$\langle V_{ee} \rangle = \left(\frac{e^2}{4\pi\epsilon_0} \right) \left(\frac{8}{\pi a^3} \right)^2 \int \frac{e^{-4(r_1+r_2)/a}}{|\mathbf{r}_1 - \mathbf{r}_2|} d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad [7.20]$$

³There do exist exactly soluble three-body problems with many of the qualitative features of helium, but using non-Coulombic potentials (see Problem 7.15).

⁴Here a is the ordinary Bohr radius and $E_n = -13.6/n^2$ eV is the n th Bohr energy; recall that for a nucleus with atomic number Z , $E_n \rightarrow Z^2 E_n$ and $a \rightarrow a/Z$ (Problem 4.17). The spin configuration associated with Equation 7.17 will be antisymmetric (the singlet).

⁵You can, if you like, interpret Equation 7.19 as first-order perturbation theory, with V_{ee} as H' . However, I regard this as a misuse of the method, since the perturbation is roughly equal in size to the unperturbed potential. I prefer, therefore, to think of it as a variational calculation, in which we are looking for an upper bound on E_g .

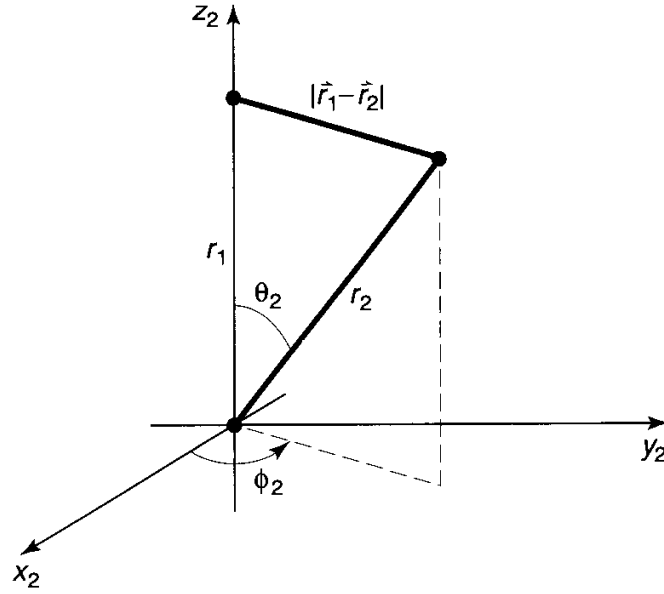


Figure 7.4: Choice of coordinates for the $\mathbf{r}_2$ integral (Equation 7.20).

I'll do the $\mathbf{r}_2$ integral first; for this purpose $\mathbf{r}_1$ is fixed, and we may as well orient the $\mathbf{r}_2$ coordinate system so that the polar axis lies along $\mathbf{r}_1$ (see Figure 7.4). By the law of cosines,

$$|\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2}, \quad [7.21]$$

and hence

$$I_2 \equiv \int \frac{e^{-4r_2/a}}{|\mathbf{r}_1 - \mathbf{r}_2|} d^3\mathbf{r}_2 = \int \frac{e^{-4r_2/a}}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2}} r_2^2 \sin \theta_2 dr_2 d\theta_2 d\phi_2. \quad [7.22]$$

The ϕ_2 integral is trivial (2π); the θ_2 integral is

$$\begin{aligned} \int_0^\pi \frac{\sin \theta_2}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2}} d\theta_2 &= \left. \frac{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_2}}{r_1r_2} \right|_0^\pi \\ &= \frac{1}{r_1r_2} \left(\sqrt{r_1^2 + r_2^2 + 2r_1r_2} - \sqrt{r_1^2 + r_2^2 - 2r_1r_2} \right) \\ &= \frac{1}{r_1r_2} [(r_1 + r_2) - |r_1 - r_2|] = \begin{cases} 2/r_1, & \text{if } r_2 < r_1, \\ 2/r_2, & \text{if } r_2 > r_1. \end{cases} \end{aligned} \quad [7.23]$$

Thus

$$\begin{aligned} I_2 &= 4\pi \left(\frac{1}{r_1} \int_0^{r_1} e^{-4r_2/a} r_2^2 dr_2 + \int_{r_1}^\infty e^{-4r_2/a} r_2 dr_2 \right) \\ &= \frac{\pi a^3}{8r_1} \left[1 - \left(1 + \frac{2r_1}{a} \right) e^{-4r_1/a} \right]. \end{aligned} \quad [7.24]$$

It follows that $\langle V_{ee} \rangle$ is equal to

$$\left(\frac{e^2}{4\pi\epsilon_0} \right) \left(\frac{8}{\pi a^3} \right) \int \left[1 - \left(1 + \frac{2r_1}{a} \right) e^{-4r_1/a} \right] e^{-4r_1/a} r_1 \sin \theta_1 dr_1 d\theta_1 d\phi_1.$$

The angular integrals are easy (4π), and the r_1 integral becomes

$$\int_0^\infty \left[r e^{-4r/a} - \left(r + \frac{2r^2}{a} \right) e^{-8r/a} \right] dr = \frac{5a^2}{128}.$$

Finally, then,

$$\langle V_{ee} \rangle = \frac{5}{4a} \left(\frac{e^2}{4\pi\epsilon_0} \right) = -\frac{5}{2} E_1 = 34 \text{ eV}, \quad [7.25]$$

and therefore

$$\langle H \rangle = -109 \text{ eV} + 34 \text{ eV} = -75 \text{ eV}. \quad [7.26]$$

Not bad (remember, the experimental value is -79 eV). But we can do better.

Can we think of a more realistic trial function than ψ_0 (which treats the two electrons as though they did not interact at all)? Rather than completely *ignoring* the influence of the other electron, let us say that, on the average, each electron represents a cloud of negative charge which partially *shields* the nucleus, so that the other electron actually sees an *effective* nuclear charge (Z) that is somewhat *less* than 2. This suggests that we use a trial function of the form

$$\psi_1(\mathbf{r}_1, \mathbf{r}_2) \equiv \frac{Z^3}{\pi a^3} e^{-Z(r_1+r_2)/a}. \quad [7.27]$$

We'll treat Z as a variational parameter, picking the value that minimizes $\langle H \rangle$.

This wave function is an eigenstate of the "unperturbed" Hamiltonian (neglecting electron repulsion), but with Z , instead of 2, in the Coulomb terms. With this in mind, we rewrite H (Equation 7.14) as follows:

$$\begin{aligned} H = & -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\epsilon_0} \left(\frac{Z}{r_1} + \frac{Z}{r_2} \right) \\ & + \frac{e^2}{4\pi\epsilon_0} \left(\frac{(Z-2)}{r_1} + \frac{(Z-2)}{r_2} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right). \end{aligned} \quad [7.28]$$

The expectation value of H is evidently

$$\langle H \rangle = 2Z^2 E_1 + 2(Z-2) \left(\frac{e^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle + \langle V_{ee} \rangle. \quad [7.29]$$

Here $\langle 1/r \rangle$ is the expectation value of $1/r$ in the (one-particle) hydrogenic ground state ψ_{100} (but with nuclear charge Z); according to Equation 6.54,

$$\left\langle \frac{1}{r} \right\rangle = \frac{Z}{a}. \quad [7.30]$$

The expectation value of V_{ee} is the same as before (Equation 7.25), except that instead of $Z = 2$ we now want *arbitrary* Z —so we multiply a by $2/Z$:

$$\langle V_{ee} \rangle = \frac{5Z}{8a} \left(\frac{e^2}{4\pi\epsilon_0} \right) = -\frac{5Z}{4} E_1. \quad [7.31]$$

Putting all this together, we find

$$\langle H \rangle = [2Z^2 - 4Z(Z - 2) - (5/4)Z] E_1 = [-2Z^2 + (27/4)Z] E_1. \quad [7.32]$$

According to the variational principle, this quantity exceeds E_g for *any* value of Z . The *lowest* upper bound occurs when $\langle H \rangle$ is minimized:

$$\frac{d}{dZ} \langle H \rangle = [-4Z + (27/4)] E_1 = 0,$$

from which it follows that

$$Z = \frac{27}{16} = 1.69. \quad [7.33]$$

This is a reasonable result; it tells us that the other electron partially screens the nucleus, reducing its effective charge from 2 down to 1.69. Putting in this value for Z , we find

$$\langle H \rangle = \frac{1}{2} \left(\frac{3}{2} \right)^6 E_1 = -77.5 \text{ eV}. \quad [7.34]$$

The ground state of helium has been calculated with great precision in this way, using increasingly complicated trial wave functions with more and more adjustable parameters.⁶ But we're within 2% of the correct answer, and, frankly, at this point my own interest in the problem begins to fade.

Problem 7.6 Using $E_g = -79.0$ eV for the ground-state energy of helium, calculate the ionization energy (the energy required to remove just *one* electron). *Hint:* First calculate the ground-state energy of the helium ion, He^+ , with a single electron orbiting the nucleus; then subtract the two energies.

***Problem 7.7** Apply the techniques of this Section to the H^- and Li^+ ions (each has two electrons, like helium, but nuclear charges $Z = 1$ and $Z = 3$, respectively). Find the effective (partially shielded) nuclear charge, and determine the best upper bound on E_g , for each case. *Note:* In the case of H^- you should find that $\langle H \rangle > -13.6$ eV, which would appear to indicate that there is no bound state at all, since it is energetically favorable for one electron to fly off, leaving behind a neutral hydrogen atom. This is not entirely surprising, since the electrons are less strongly attracted to the nucleus than they are in helium, and the electron repulsion tends to break the atom apart. However, it turns out to be incorrect. With a more sophisticated trial wave

⁶E. A. Hylleraas, *Z. Phys.* **65**, 209 (1930); C. L. Pekeris, *Phys. Rev.* **115**, 1216 (1959).

function (see Problem 7.16) it can be shown that $E_g < -13.6$ eV, and hence that a bound state does exist. It's only *barely* bound, however, and there are no excited bound states,⁷ so H^- has no discrete spectrum (all transitions are to and from the continuum). As a result, it is difficult to study in the laboratory, although it exists in great abundance on the surface of the sun.⁸

7.3 THE HYDROGEN MOLECULE ION

Another classic application of the variational principle is to the hydrogen molecule ion, H_2^+ , consisting of a single electron in the Coulomb field of two protons (Figure 7.5). We shall assume for the moment that the protons are fixed in position, a specified distance R apart, although one of the most interesting byproducts of the calculation is going to be the actual *value* of R . The Hamiltonian is

$$H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} \right), \quad [7.35]$$

where r_1 and r_2 are the distances to the electron from the respective protons. As always, the strategy will be to guess a reasonable trial wave function, and invoke the variational principle to get a bound on the ground-state energy. (Actually, our main interest is in finding out whether this system bonds at *all*—that is, whether its energy is less than that of a neutral hydrogen atom plus a free proton. If our trial wave function indicates that there *is* a bound state, a *better* trial function can only make the bonding even stronger.)

To construct the trial wave function, imagine that the ion is formed by taking a hydrogen atom in its ground state (Equation 4.80),

$$\psi_g(\mathbf{r}) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}, \quad [7.36]$$

and then bringing in a proton from far away and nailing it down a distance R away. If R is substantially greater than the Bohr radius, a , the electron's wave function

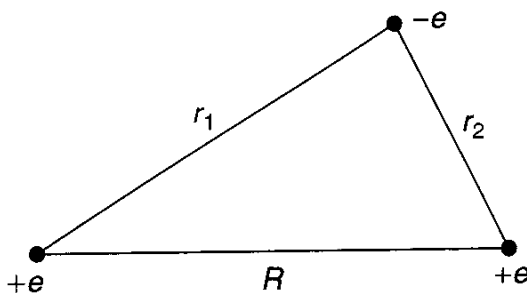


Figure 7.5: The hydrogen molecule ion, H_2^+ .

⁷Robert N. Hill, *J. Math. Phys.* **18**, 2316 (1977).

⁸For further discussion, see Hans A. Bethe and Edwin E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms* (New York: Plenum 1977), Section 34.

probably isn't changed very much. But we would like to treat the two protons on an equal footing, so that the electron has the same probability of being associated with either one. This suggests that we consider a trial function of the form

$$\psi = A [\psi_g(r_1) + \psi_g(r_2)]. \quad [7.37]$$

(Quantum chemists call this the **LCAO** technique, because we are expressing the *molecular* wave function as a *linear combination of atomic orbitals*.)

Our first task is to *normalize* the trial function:

$$\begin{aligned} 1 = \int |\psi|^2 d^3\mathbf{r} &= |A|^2 \left[\int |\psi_g(r_1)|^2 d^3\mathbf{r} \right. \\ &\quad \left. + \int |\psi_g(r_2)|^2 d^3\mathbf{r} + 2 \int \psi_g(r_1) \psi_g(r_2) d^3\mathbf{r} \right]. \end{aligned} \quad [7.38]$$

The first two integrals are 1 (since ψ_g itself is normalized); the third is more tricky. Let

$$I \equiv \langle \psi_g(r_1) | \psi_g(r_2) \rangle = \frac{1}{\pi a^3} \int e^{-(r_1+r_2)/a} d^3\mathbf{r}. \quad [7.39]$$

Picking coordinates so that proton 1 is at the origin and proton 2 is on the z -axis at the point R (Figure 7.6), we have

$$r_1 = r \text{ and } r_2 = \sqrt{r^2 + R^2 - 2rR \cos \theta}, \quad [7.40]$$

and therefore

$$I = \frac{1}{\pi a^3} \int e^{-r/a} e^{-\sqrt{r^2 + R^2 - 2rR \cos \theta}/a} r^2 \sin \theta dr d\theta d\phi. \quad [7.41]$$

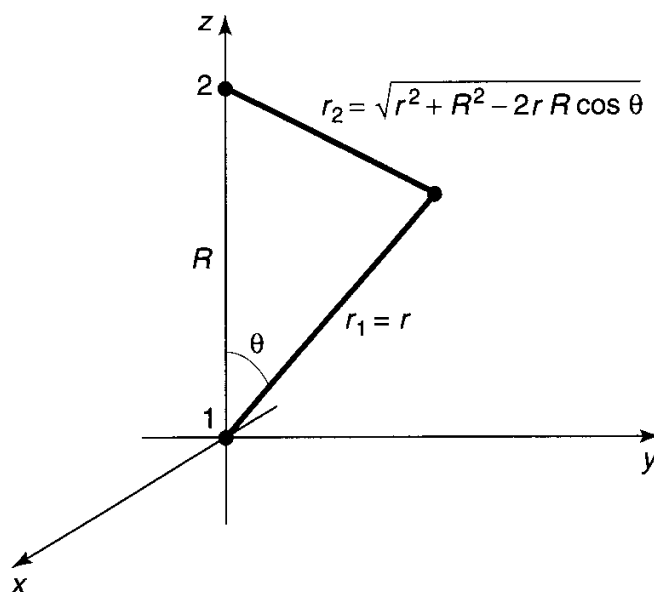


Figure 7.6: Coordinates for the calculation of I (Equation 7.39).

The ϕ integral is trivial (2π). To do the θ integral, let

$$y \equiv \sqrt{r^2 + R^2 - 2rR \cos \theta}, \text{ so that } d(y^2) = 2y dy = 2rR \sin \theta d\theta.$$

Then

$$\begin{aligned} \int_0^\pi e^{-\sqrt{r^2 + R^2 - 2rR \cos \theta}/a} \sin \theta d\theta &= \frac{1}{rR} \int_{|r-R|}^{r+R} e^{-y/a} y dy \\ &= -\frac{a}{rR} \left[e^{-(r+R)/a} (r+R+a) - e^{-|r-R|/a} (|r-R|+a) \right]. \end{aligned}$$

The r integral is now straightforward:

$$\begin{aligned} I &= \frac{2}{a^2 R} \left[-e^{-R/a} \int_0^\infty (r+R+a) e^{-2r/a} r dr + e^{-R/a} \int_0^R (R-r+a) r dr \right. \\ &\quad \left. + e^{R/a} \int_R^\infty (r-R+a) e^{-2r/a} r dr \right]. \end{aligned}$$

Evaluating the integrals, we find (after some algebraic simplification),

$$I = e^{-R/a} \left[1 + \left(\frac{R}{a} \right) + \frac{1}{3} \left(\frac{R}{a} \right)^2 \right]. \quad [7.42]$$

I is called an **overlap** integral; it measures the amount by which $\psi_g(r_1)$ overlaps $\psi_g(r_2)$ (notice that it goes to 1 as $R \rightarrow 0$, and to 0 as $R \rightarrow \infty$). In terms of I , the normalization factor (Equation 7.38) is

$$|A|^2 = \frac{1}{2(1+I)}. \quad [7.43]$$

Next we must calculate the expectation value of H in the trial state ψ . Noting that

$$\left(-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_1} \right) \psi_g(r_1) = E_1 \psi_g(r_1)$$

(where $E_1 = -13.6$ eV is the ground-state energy of atomic hydrogen)—and the same with r_2 in place of r_1 —we have

$$\begin{aligned} H\psi &= A \left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \right] [\psi_g(r_1) + \psi_g(r_2)] \\ &= E_1 \psi - A \left(\frac{e^2}{4\pi\epsilon_0} \right) \left[\frac{1}{r_2} \psi_g(r_1) + \frac{1}{r_1} \psi_g(r_2) \right]. \end{aligned}$$

It follows that

$$\langle H \rangle = E_1 - 2|A|^2 \left(\frac{e^2}{4\pi\epsilon_0} \right) \left[\langle \psi_g(r_1) | \frac{1}{r_2} | \psi_g(r_1) \rangle + \langle \psi_g(r_1) | \frac{1}{r_1} | \psi_g(r_2) \rangle \right]. \quad [7.44]$$

I'll let you calculate the two remaining quantities, the so-called **direct integral**,

$$D \equiv a \left\langle \psi_g(r_1) \left| \frac{1}{r_2} \right| \psi_g(r_1) \right\rangle, \quad [7.45]$$

and the **exchange integral**,

$$X \equiv a \left\langle \psi_g(r_1) \left| \frac{1}{r_1} \right| \psi_g(r_2) \right\rangle. \quad [7.46]$$

The results (see Problem 7.8) are

$$D = \frac{a}{R} - \left(1 + \frac{a}{R}\right) e^{-2R/a} \quad [7.47]$$

and

$$X = \left(1 + \frac{R}{a}\right) e^{-R/a}. \quad [7.48]$$

Putting all this together, and recalling (Equations 4.70 and 4.72) that $E_1 = -(e^2/4\pi\epsilon_0)(1/2a)$, we conclude that

$$\langle H \rangle = \left[1 + 2 \frac{(D + X)}{(1 + I)} \right] E_1. \quad [7.49]$$

According to the variational principle, the ground-state energy is *less* than $\langle H \rangle$. Of course, this is only the *electron's* energy—there is also potential energy associated with the proton-proton repulsion:

$$V_{pp} = \frac{e^2}{4\pi\epsilon_0} \frac{1}{R} = -\frac{2a}{R} E_1. \quad [7.50]$$

Thus the *total* energy of the system, in units of $-E_1$ and expressed as a function of $x \equiv R/a$, is less than

$$F(x) = -1 + \frac{2}{x} \left\{ \frac{(1 - (2/3)x^2)e^{-x} + (1 + x)e^{-2x}}{1 + (1 + x + (1/3)x^2)e^{-x}} \right\}. \quad [7.51]$$

This function is plotted in Figure 7.7. Evidently bonding *does* occur, for there exists a region in which the graph goes below -1 , indicating that the energy is less than that of a neutral atom plus a free proton (to wit, -13.6 eV). The equilibrium separation of the protons is about 2.4 Bohr radii, or 1.27 Å.

***Problem 7.8** Evaluate D and X (Equations 7.45 and 7.46). Check your answers against Equations 7.47 and 7.48.

****Problem 7.9** Suppose we used a *minus* sign in our trial wave function (Equation 7.37):

$$\psi = A[\psi_g(r_1) - \psi_g(r_2)]. \quad [7.52]$$

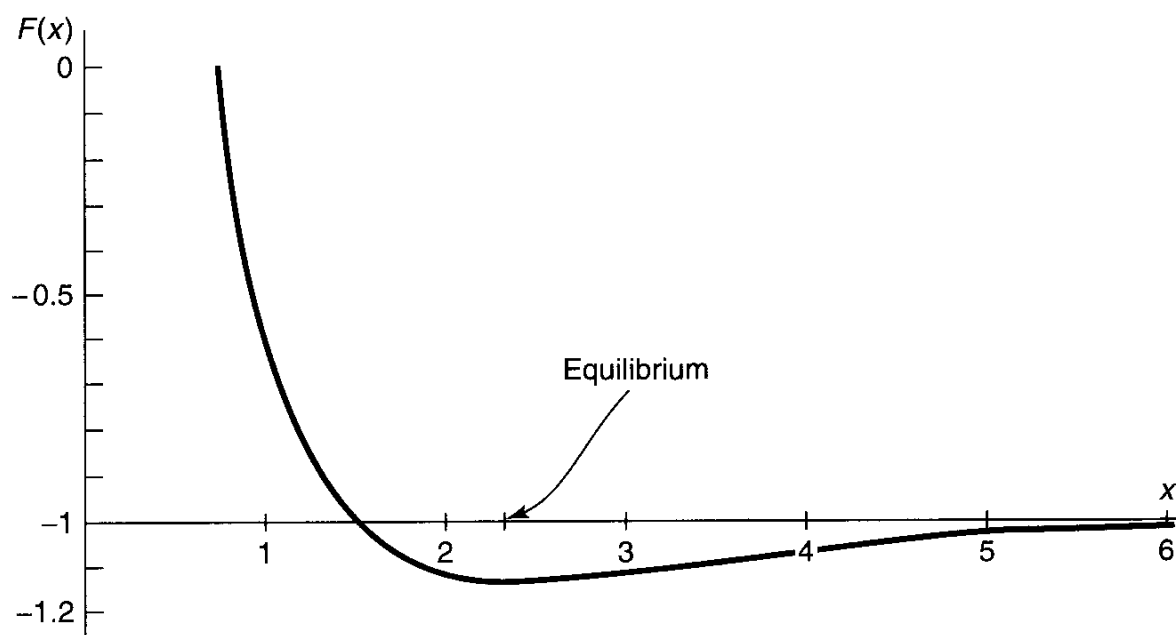


Figure 7.7: Plot of the function $F(x)$, Equation 7.51, showing existence of a bound state.

Without doing any new integrals, find $F(x)$ (the analog to Equation 7.51) for this case, and construct the graph. Show that there is no evidence of bonding. (Since the variational principle only gives an *upper bound*, this doesn't *prove* that bonding cannot occur for such a state, but it certainly doesn't look promising). *Note:* Actually, any function of the form

$$\psi = A[\psi_g(r_1) + e^{i\phi}\psi_g(r_2)] \quad [7.53]$$

has the desired property that the electron is equally likely to be associated with either proton. However, since the Hamiltonian (Equation 7.35) is invariant under the interchange $P:r_1 \leftrightarrow r_2$, its eigenfunctions can be chosen to be simultaneously eigenfunctions of P . The plus sign (Equation 7.37) goes with the eigenvalue $+1$, and the minus sign (Equation 7.52) with the eigenvalue -1 ; nothing is to be gained by considering the ostensibly more general case (Equation 7.53), though you're welcome to try it, if you're interested.

*****Problem 7.10** The second derivative of $F(x)$, at the equilibrium point, can be used to estimate the natural frequency of vibration (ω) of the two protons in the hydrogen molecule ion (see Section 2.3). If the ground-state energy ($\hbar\omega/2$) of this oscillator exceeds the binding energy of the system, it will fly apart. Show that in fact the oscillator energy is small enough that this will not happen, and estimate how many bound vibrational levels there are. *Note:* You're not going to be able to obtain the position of the minimum—still less the second derivative at that point—analytically. Do it numerically, on a computer.

FURTHER PROBLEMS FOR CHAPTER 7

Problem 7.11 Find the lowest bound on the ground state of hydrogen you can get using a gaussian trial wave function

$$\psi(\mathbf{r}) = Ae^{-br^2},$$

where A is determined by normalization and b is an adjustable parameter. *Answer:* -11.5 eV.

****Problem 7.12** If the photon had a nonzero mass ($m_\gamma \neq 0$), the Coulomb potential would be replaced by a **Yukawa potential**, of the form

$$V(\mathbf{r}) = -\frac{e^2}{4\pi\epsilon_0} \frac{e^{-\mu r}}{r}, \quad [7.54]$$

where $\mu = m_\gamma c/\hbar$. With a trial wave function of your own devising, estimate the binding energy of a “hydrogen” atom with this potential. Assume $\mu a \ll 1$, and give your answer correct to order $(\mu a)^2$.

Problem 7.13 Suppose you’re given a quantum system whose Hamiltonian H_0 admits just two eigenstates, ψ_a (with energy E_a), and ψ_b (with energy E_b). They are orthogonal, normalized, and nondegenerate (assume E_a is the smaller of the two). Now we turn on a perturbation H' , with the following matrix elements:

$$\langle \psi_a | H' | \psi_a \rangle = \langle \psi_b | H' | \psi_b \rangle = 0; \quad \langle \psi_a | H' | \psi_b \rangle = \langle \psi_b | H' | \psi_a \rangle = h. \quad [7.55]$$

- Find the exact eigenvalues of the perturbed Hamiltonian.
- Estimate the energies of the perturbed system using second-order perturbation theory.
- Estimate the ground-state energy of the perturbed system using the variational principle, with a trial function of the form

$$\psi = (\cos \phi) \psi_a + (\sin \phi) \psi_b, \quad [7.56]$$

where ϕ is an adjustable parameter. (Note that writing the linear combination in this way guarantees that ψ is normalized.)

- Compare your answers to (a), (b), and (c). Why is the variational principle so accurate in this case?

Problem 7.14 As an explicit example of the method developed in Problem 7.13, consider an electron at rest in a uniform magnetic field $\mathbf{B} = B_z \hat{k}$, for which the Hamiltonian is (Equation 4.158):

$$H_0 = \frac{eB_z}{m} S_z. \quad [7.57]$$

The eigenspinors, χ_a and χ_b , and the corresponding energies, E_a and E_b , are given in Equation 4.161. Now we turn on a perturbation, in the form of a uniform field in the x direction:

$$H' = \frac{eB_x}{m} S_x. \quad [7.58]$$

- (a) Find the matrix elements of H' , and confirm that they have the structure of Equation 7.55. What is h ?
- (b) Using your result in Problem 7.13(b), find the new ground-state energy, in second-order perturbation theory.
- (c) Using your result in Problem 7.13(c), find the variational principle bound on the ground-state energy.

*****Problem 7.15** Although the Schrödinger equation for helium itself cannot be solved exactly, there exist “helium like” systems that do admit exact solutions. A simple example⁹ is “rubber-band helium”, in which the Coulomb forces are replaced by Hooke’s law forces:

$$H = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) + \frac{1}{2}m\omega^2(r_1^2 + r_2^2) - \frac{\lambda}{4}m\omega^2|\mathbf{r}_1 - \mathbf{r}_2|^2. \quad [7.59]$$

- (a) Show that the change of variables from $\mathbf{r}_1, \mathbf{r}_2$, to

$$\mathbf{u} \equiv \frac{1}{\sqrt{2}}(\mathbf{r}_1 + \mathbf{r}_2), \quad \mathbf{v} \equiv \frac{1}{\sqrt{2}}(\mathbf{r}_1 - \mathbf{r}_2) \quad [7.60]$$

turns the Hamiltonian into two independent three-dimensional harmonic oscillators:

$$H = \left[-\frac{\hbar^2}{2m}\nabla_u^2 + \frac{1}{2}m\omega^2 u^2 \right] + \left[-\frac{\hbar^2}{2m}\nabla_v^2 + \frac{1}{2}(1-\lambda)m\omega^2 v^2 \right]. \quad [7.61]$$

- (b) What is the *exact* ground-state energy for this system?
- (c) If we didn’t know the exact solution, we might be inclined to apply the method of Section 7.2 to the Hamiltonian in its original form (Equation 7.59). Do so (but don’t bother with shielding). How does your result compare with the exact answer? *Answer:* $\langle H \rangle = 3\hbar\omega(1 - \lambda/4)$.

*****Problem 7.16** In Problem 7.7 we found that the trial wave function with shielding (Equation 7.27), which worked well for helium, is inadequate to confirm the existence

⁹For a more sophisticated model, see R. Crandall, R. Whinnell, and R. Bettega, *Am. J. Phys.* **52**, 438 (1984).

of a bound state for the negative hydrogen ion. Chandrasekhar¹⁰ used a trial wave function of the form

$$\psi(\mathbf{r}_1, \mathbf{r}_2) \equiv A[\psi_1(r_1)\psi_2(r_2) + \psi_2(r_1)\psi_1(r_2)], \quad [7.62]$$

where

$$\psi_1(r) \equiv \sqrt{\frac{Z_1^3}{\pi a^3}} e^{-Z_1 r/a}, \text{ and } \psi_2(r) \equiv \sqrt{\frac{Z_2^3}{\pi a^3}} e^{-Z_2 r/a}. \quad [7.63]$$

In effect, he allowed two *different* shielding factors, suggesting that one electron is relatively close to the nucleus and the other is farther out. (Because electrons are identical particles, the spatial wave function must be symmetrized with respect to interchange. The *spin* state—which is irrelevant to the calculation—is evidently antisymmetric.) Show that by astute choice of the adjustable parameters Z_1 and Z_2 you can get $\langle H \rangle$ less than -13.6 eV. *Answer:*

$$\langle H \rangle = \frac{E_1}{x^6 + y^6} \left(-x^8 + 2x^7 + \frac{1}{2}x^6y^2 - \frac{1}{2}x^5y^2 - \frac{1}{8}x^3y^4 + \frac{11}{8}xy^6 - \frac{1}{2}y^8 \right),$$

where $x \equiv Z_1 + Z_2$ and $y \equiv 2\sqrt{Z_1 Z_2}$. Chandrasekhar used $Z_1 = 1.039$ (since this is larger than 1, the motivating interpretation as an effective nuclear charge cannot be sustained, but never mind—it's still an acceptable trial wave function) and $Z_2 = 0.283$.

Problem 7.17 The fundamental problem in harnessing nuclear fusion is getting the two particles (say, two deuterons) close enough together for the attractive (but short-range) nuclear force to overcome the Coulomb repulsion. The “brute force” method is to heat the particles to fantastic temperatures and allow the random collisions to bring them together. A more exotic proposal is **muon catalysis**, in which we construct a “hydrogen molecule ion”, only with deuterons in place of protons, and a *muon* in place of the electron. Predict the equilibrium separation distance between the deuterons in such a structure, and explain why muons are superior to electrons for this purpose.¹¹

¹⁰S. Chandrasekhar, *Astrophys. J.* **100**, 176 (1944).

¹¹The classic paper on muon-catalyzed fusion is J. D. Jackson, *Phys. Rev.* **106**, 330 (1957); for a recent popular review, see J. Rafelski and S. Jones, *Scientific American*, November 1987, page 84.